NYSE Euronext

WSE

XDP CDE Message Specifications

Version 6.8 - 11 February 2020

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PREFACE

Document Purpose

This document sets out the XDP message specifications for the cash, derivatives and ETP markets.

Target Audience

This document should be read by WSE. This document is dedicated to members of WSE, ISV and Data Vendors willing to develop an interface to subscribe to WSE UTP market data.

Associated Documents

WSE – XDP CDE Configuration

Navigation Used Within This Document

Hyperlinks let you navigate to sections of the document, as well as between definitions and field descriptions.

- Click the hyperlinks in any part of the document to view some definitions.
- To return to the hyperlink, press [ALT] + [←] (the back/left/previous arrow).

Conventions Used Within This Document

N/A

Note : All FILLERs in XDP messages are filled with binary NULLs.

DOCUMENT HISTORY

Version	Date	Author	Description	Status
1.0	20/10/2010	LBR	Initial version.	Delivered
1.1	25/11/2010	LBR	- Miscellaneous corrections - Addition of message 561 session summary - Addition of the BondSpot messages 601, 602 and 603 - Referential message is renamed into 556	Delivered
1.2	09/12/2010	LBR	- Removal of non functional specifications - Addition of message Trade Publication 243 - Addition of generic WSE message 592	Delivered
1.3	05/01/2011	LBR	- Miscellaneous corrections to message 243 - Addition of the Settlement Delay to message 556	Delivered
1.4	17/01/2011	LBR	- QuoteLinkID is not used in messages 220 and 240 - Removal of field SessionType in message 539 - Change of values for field Side in message 142	Delivered
1.5	24/02/2011	LBR	- 505/516: relationship between InstrumentState and ClassState - Addition of SymbolIndex to messages 561, 601, 602 and 603 - 516: SessionType values E and L are not provided - Addition of SystemID in messages 601, 602 and 603 - Message 560 is renamed Short Sale Change	Delivered
1.6	26/04/2011	LBR	- Adjustment of the corporate action types list - Addition of details to the cumulative quantity in message 245 - InstrumentGroupCode length is 3 instead of 2 in messages 561, 601 and 602 - Addition of a filler to message 603	Delivered
1.7	28/06/2011	LBR	UTP-CDE release - New values added to field TypeOfPrice in message 241 - New values added to fields InstrumentCategory, MarketSegment in message 556	Delivered
			- New fields OptionType, DeliveryType, ExerciseType, WarrantThresholdMin, WarrantThresholdMinScaleCode, WarrantThresholdMax, WarrantThresholdMaxScaleCode added to message 556 - Field UnderlyingWISINCode is renamed into UnderlyingISINCode in message 556 - Addition of open position message 595 - New values added to fields HaltReason, ActionAffectingState, PeriodSide in message 505 - Clarification on headers (dedicated section)	
1.8	03/08/2011	ALX LBR	- Clarification on Offsets - Message size adjustment (fillers and trailing fillers): . 539 Session TimeTable . 561 Session Summary . 220 Trade Creation . 240 Trade Full information - New indices messages: . 546 Real time index . 547 Index Summary . 548 Index Portfolio - New fields MinQuantity, MaxQuantity, Multiplier, MultiplierScaleCode added to message 556	Delivered
1.8a	27/09/2011	LBR	- Update of the message sending rules for message 505 - Field RepoExpiryDate is renamed into ExpiryDate in message 556 - Addition of the following MIC 'BOSP' - BondSpot MTF 'RPWC' - BondSpot Regulated Market 'TBSP' - Treasury BondSpot Poland	Delivered
1.9	20/10/2011	FLO WSE	- Addition of fields SourceSeqNum, SourceTime, SystemID and SourceTimeMicroSecs in messages 556, 560, 561, 595, 221, 247, 142, 231, 546, 547, 548, 601, 602, 603 - Addition of descriptions for BondSpot	Delivered
1.9a	18/11/2011	FLO	- Clarification on Market Sheet sequencing - Addition of values 5-Internalized Trade' and '6-Internalized Cross Trade' in XDP240 "TradeCond3" description - Addition of value '4-Derivatives' in XDP523 "Scope" description	Delivered

Version	Date	Author	Description	Status
1.9b	02/12/2011	FLO	- Addition of format type for Accrued Interest group field in message 601; - Addition of a note regarding message 220	Delivered
1.9c	14/12/2011	ALX	- Message 140. Modification of Best Limits from BB010 to BB05	Delivered
2.0	06/01/2012	FLO	- Message 505 : update (Field InstrumentTrading is not used) - Message 505 : Addition of Instrument Change parameters matrix - Message 556 : OfficialQuotationList, Heading Number and Section number are replaced by one Filler of 3 chars at position 400 - Message 556 : Addition of a Filler of 2 chars at position 494 - Message 556 : offset correction for last few fields (starting at WarrantThresholdMin) - Message 602 : SettlementDate field is now 8 char.	Delivered
2.0a	21/02/2012	FLO	- §1.3.3 Price Update (MDD 241 message): Correction in the TypeOfPrice value range. Authorized values are: 02, 03, 33, 34, 35, 36, 37 and 51 - §1.4 Quote and BB05: Addition of a note on SourceSeqNum - §1.2.5 Indicative Matching Price : Addition of message sending rules	Delivered
2.0b	27/02/2012	FLO	- Message 556 : field SmallTrade at position 480 is renamed BlockSize and is used to disseminate the instrument normal block size	Delivered
2.0c	06/03/2012	FLO	- Date format is YYYYMMDD for fields : - Message 556 : EventDate, ExpiryDate - Message 595 : TradingDate - Message 243 : TradingDate, ReportingDate - Message 548 : DateValidity - Message 505 : field PeriodSide is equal to '1' for Bid Only and '2' for Offer Only	Delivered
2.1	15/03/2012	FLO	- Message 241 (§1.3.3): Addition of value 30 (First Trade of the day) in TypeOfPrice value range.	Delivered
2.1a	22/03/2012	FLO	- Message 556 (§1.2.10): Field QuantityNotation is replaced by a Filler at position 204.	Delivered
2.2	12/04/2012	FLO	- Message XDP220 has been decommissioned - Clarification on messages : 556, 561, 595, 240, 240, 245, 140, 230, 546, 547 - Format of Fillers	Delivered
			Addition of Appendices §2.4 Sectors, §2.5 Segments, §2.6 Indices - Message 548 :field NonTransPricesMarker (position 697) is replaced by a filler - Message 505 (§1.2.2): Clarification on message sending rules. - Message 530 (§1.2.51.2.2): Clarification on message sending rules.	
2.3	06/06/2012	FLO	- Addition of appendix MIC - Message 243 : field 'Venue' (offset130) has been renamed 'MIC' - Addition of detail structure for WSE generic message - Message 505 : value 'G' has been removed from the value range of the field HaltReason - Message 140 : Clarification on fields NumberAskOrders and NumberBidOrders	Delivered
2.4	05/09/2012	FLO	-Value 'P' has been removed from ActionType field range - Update in BondSpot value range of field TypeOfPrice -The value IndexLevel has been added to field LevelScaleCode - Column 'Number of Field' has been removed from YTM Appendix - Clarification on TradingTime and ReportingTime fields -Updates in Stock Types Appendix	Delivered
2.5	14/09/2012	FLO	- Stock Types Appendix update	Delivered
2.6	21/09/2012	FLO	- MIC Codes Appendix update - Trade Publication - 243 Message: field at position 130 is restored 'Venue' -Order Update / Market Sheet - 230 Message: Market sheet sequencing at Continuous phase is Market first, then Limits AND Pegs (with the same level of priority) -Session Summary - 561 Message: Clarification on SettlementValue field	Delivered
2.7	01/11/2012	FLO	-Instrument State Change Matrix: Restoration of value 'G' and addition of value 'S' in field HaltReason	Delivered

Version	Date	Author	Description	Status
			-VWAP – Closing Price - 247 Message: values ClosingPriceRule, ClosingPrice, NbTrades, QtyShares and AmountTraded are not provided for derivatives instruments	
2.8	10/10/2012	FLO	-Session Summary - 561 Message: inversion of TransactionsNumber and TradingValue fields, extension of TradingValue and TradingValueCurrency fields length to 8 Binary Int. - Real time Index - 546 Message: extension of ValueTrading field length to 8 Binary Int. -Price Update - 241 Message: clarification in message sending rule (type of price 34)	Delivered
2.9	17/10/2012	FLO	-VWAP – Closing Price - 247 Message: rename PriceScaleCode to ScaleCode -Addition of MIC XXXX -Instrument State Change Matrix update in InstrumentState and ActionAffectingState values for unhalt instrument cases (unhalted by Market Operations in Core Continuous phase & Auction by Market Operations)	Delivered
3.0	19/10/2012	FLO	-VWAP – Closing Price - 247 Message: extension of field AmountTraded to 8 characters - The sequence between fields AmountTraded and QtyShares has been changed - The filler at position 51 has been extended -Session Summary - 561 Message: clarification in message overview	Delivered
4.0	10/12/2012	FLO	-Generic WSE Message - 592 Message: Correction in the name of field TypeOfRecipient (instead of TypeOffRecipient) -Instrument State Change Matrix: The value of HaltReason when an instrument has been halted by the system in Core Continuous due to the breaching of a static threshold is 'S' -Referential - 556 Message: Addition of Blank value for fields SSMarketMaker and SSNonMarketMaker (meaning that those two fields are not significant for warrants and derivatives instruments, and instruments traded on IPO and TO segements) -MIC Codes: Addition of MIC Code WBCL (Warsaw Stock Exchange/ Bonds/Catalyst/Listing) -Stock Types	Delivered
			Addition of stock types 208 (Mortgage-Backed Bonds) and 209 (Public Mortgage Bonds) -Referential - 556 Message: Correction of the name of the field 'NomMkdtPriceScaleCode' to 'NomMktPriceScaleCode' Correction of descrpiton of the field: 'NomMkdtPrice' to 'NomMktPrice'	
4.1	15/01/2013	FLO	-Order Book Retransmission Delimiter - 231 Message: clarification on fields TradingEngineID (which is renamed ApplicationID) and InstanceID	Delivered
4.2	28/01/2013	FLO	-Instrument State Change - 505 Message: addition of value 'A' in Haltreason value range	Delivered
4.3	12/02/2013	FLO	-Instrument State Change Matrix: clarification in two unhalt cases (instrument unhalted by MO, Auction by MO) and in one halt case (Trading Engine Initialization) -Instrument State Change - 505 Message: removal of value 'C' from field InstrumentState value range -Session Summary - 561 Message: In TradingVolume field description, the comment "For instruments traded in currency other than PLN, value is converted on PLN according to currency price given in field CurrencyPrice" has been be removed -VWAP – Closing Price - 247 Message: add a comment in ScaleCode field description saying that it is also applicable to AmountTraded. -Order Update / Market Sheet - 230 Message: Correction in ActionType field: 'D' - Deletion of order identified by OrderID or Deletion of all orders for the given instrument (depending on the side. If side is not provided, it means both) 'F' - Deletion all orders for one instrument at the start of the day before the order book retransmission	Delivered
4.4	26/02/2013	FLO	-Instrument State Change - 505 Message: removal of value 'A' from field InstrumentState value range. -Instrument State Change Matrix has been renamed Instrument State Change Parameters in order to avoid any confusion with MATRIX application. -Instrument State Change Parameters:	Delivered

Version	Date	Author	Description	Status
			Correction in the unhalt case “instrument Auction by MO”: value ‘A’ is never generated in field InstrumentState. -Order Update / Market Sheet - 230 Message: Restoration of description for value ‘D’ in ActionType field:‘D’ - Deletion of order identified by OrderID Interest for Settlement Day - 602 Message: addition of the following message sending rule: -During the day, each time the value of interest for settlement day, for a given bond or mortgage-backed bond is changed. - Clarification on SourceSeqNum field description in all messages: In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit. Except in message 140, SourceSeqNum increases serially and does not increase monotonically. Except in message 231 (for which there is no RFS mode): In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. -Yield to Maturity - 603 Message: addition of sendind rules in case of 140 messages occurred: ○ The 603 message is sent for each price level changing, separately for bid and ask price, in case of both are sent within 140, ○ The 603 messages are sent right after the package of 140 concerning the same event is sent.	
4.5	27/02/2013	FLO	-Order Book Retransmission Delimiter - 231 Message: field InstanceID is renamed InstanceNum: It Indicates the number of trading units for which the orderbooks are rebroadcast. (On the initial WSE configuration, the number of instances is 2 as there are 2 trading units) Valid values: 2	Delivered
4.6	21/03/2013	FLO	-Indication of Interest - 142 Message: Addition of comment in ValidityTime field description : Time use CET. -Session Timetable - 539 Message: Addition of comment in ClassStateTime field description: 0000 phase linked to the previous phase & Time use CET. -Session Summary - 561 Message: Value 'bzd' is corrected to value 'bz' in InterimDividendRight field description. -Trade Full Information - 240 Message: Addition of comment in TradeOrigin field: value ‘R’ is for future use -Real time Index - 546 Message: Addition of comment in TimeFirstHigh and TimeFirstLow fields: Time use CET. -Index Summary - 547 Message: - Addition of comment in TimeClosing, TimeOpening, TimeFullOpening, TimeHigh and TimeLow fields : Time use CET. - Addition of a comment in CapitalisationPortfolio and IndexAdjustmentCoefficient fields: Note used, For future use -Index Portfolio - 548 Message: - Addition of comment in CapitalisationPortfolio, IndexAdjustmentCoefficient field: Note used, For future use - Segments Appendix: Addition of values HR (Segment NewConnect High Liquidity Risk (HLR)) and SR (Segment NewConnectHigh Super High Liquidity Risk (SHLR)) -Trade Full Information - 240 Message : Addition of comment: Value accumulated within the individual segments	Delivered
4.7	19/07/2013	FLO	-Instrument State Change - 505 Message: restore value ‘A’ in InstrumentState field value range	Delivered

Version	Date	Author	Description	Status
			-Order Update / Market Sheet - 230 Message: In the specific case of a Peg order repricing, the sequence number remains the same (ISS116587). -Referential - 556 Message: value valeur 99 (multiple corporate Events) is added to TypeOfCorporateEvent field (ISS112373). -Generic WSE Message - 592 Message: addition of Body Message for InternalMsgType = 02. **-Indices: During the session - index status „SI", addition of indice NCIndex30 – every 5 minutes.**	
4.8	11/09/2013	FLO (mail JMA 30/08/ 2013)	-Generic WSE Message - 592 Message: **For Body Message (InternalMsgType = 02),** **1- There has been a format correction for fields Date of First Quotation (position 60) and Date (position 84) which are in ASCII format.** **2- Size correction of the Filler at the end of the body message (position 209): 59 bytes.** **-Indices: addition of index WIG30 and WIG-CEE to "index status „SI"** -Order Update / Market Sheet - 230 Message: **Addition of the following description in "OrderType" field: "Null –Not provided (case of order deletion independently of reason)"**	Delivered
4.9	20/12/2013	FLO	-Common Packet Header Format: Addition of value 18 (Retransmission message (Compressed) in DeliveryFlag field value range. -Session Summary - 561 Message: SettlementValueOpening field has been replaced by a Filler -Generic WSE Message - 592 Message: Change in message sending rules: The Generic WSE message with InternalMsgType =1 indicates the end of messages distribution for the current trading day. Message is sent at the end of the day. The Generic WSE message with InternalMsgType =2 informs about the structure of the basket of treasury bonds for given bond future for the next trading day. Message is sent during the trading day	Delivered
5.0	15/04/2014	FLO	-Session Summary - 561 Message: SettlementValueOpeningScaleCode field has been replaced by a Filler (for future use). -Referential - 556 Message: Field TypeOfCorporateEvent value '17' is for Spin-off. -Real time Index - 546 Message and Index Portfolio - 548 Message Addition of a new resolution in message sending rules : Regulations of the Management Board of the Warsaw Stock Exchange No. 871/2013 of August 6, 2013 as amended – hereinafter referred to as the WIG30, WIG50 and WIG250. -Index Summary - 547 Message: Addition of the same resolutions present in messages 546 and 547 sending rules. -Segments: Removal of segments L, M, S, B, MA, SA, BA. Update in LA segment definition. -Indices: Correction in During the session - index status „SI". „2" – "current index": Addition of WIG30short, WIG30lev, WIG50 and WIG250. Removal of sWIG80 and WIG-CEE. „B" – "synthetic opening index 100%": correction of example index WIG-Poland and removal of NCILifeScience. -Mail - 523 Message: Correction of **Scope** field value range: '1' - Warsaw Stock Exchange Markets.	Delivered

Version	Date	Author	Description	Status
5.1	10/09/2014	FLO	Instrument State Change - 505 Message: Field PeriodSide Indicates the side for an instrument beginning or ending a Bid only or Offer only situation.	Delivered
5.2	17/10/2014	FLO	-Price Update - 241 Message: changes in -Change in message sending rules for TypeOfPrice '34'. -Change in description of TypeOfPrice field for value '34'. -Session Summary - 561 Message: -Change in ClosingPrice field description -Change in StrikePrice field description -Change WarrantsTrade field name to NumberOfInstruments (+description change)	Delivered
5.3	02/12/2014	DCZ	- Real time Index - 546 Message, - Index Summary 547 Message, - Index Portfolio - 548 Message: Change in 'Message Sending Rules': Resolution of the Management Board of the Warsaw Stock Exchange No. 871/2013 of August 6, 2013 as amended – hereinafter referred to as the WIG30 index. - Indices: Change in 'During the session - index status SI': -removal of WIG50, WIG250, WIG30short and WIG30lev, -resumption of sWIG80 calculation, -change in dissemination frequency of WIG30	Delivered
5.4	18/05/2015	DCZ	- Referential - 556 Message: Addition of comment in SSMarketMaker and SSNonMarketMaker field description : Field should be ignored - Short Sale Change - 560 Message: Addition of comment: This message should be ignored - Session Summary - 561 Message: Change in ShortSellingTradesNumber, VolumeShortSellingTrades, ValueShortSellingTrades ValueShortSellingTradesScaleCode and ShortSale field description: Field should be ignored - Trade Full Information - 240 Message: Change in SSTrade field description: Filed should be ignored - Order Update/Market Sheet - 230 Message: Addition of comment in Side field description: '5' – sell short, equivalent to value '2'(sell). Use of designation '5' as 'Sell short' by Exchange Members will be optional and will not be required or monitored by the WSE. Exchange Members may use the designation to mark orders with value '5' in the 'Side' field for its own needs. - MIC Codes Removal of MIC Code WCDE and POEE	Delivered
5.5	10/06/2016	WSE	-Price Update - 241 Message: Addition of new value of field TypeOfPrice: 99 For TBSP Reference Spot Price.Update of field description 'TypeOfPrice" in case of lack of fixing price in BondSpot stream	Delivered
5.6	23/08/2016	WSE	Segments Removal of LD, HR, SR NewConnect segments. Introduction of new NF, NB, NA NewConnect segments. Indices Removal of the reference to the Resolution of the Management Board of the Warsaw Stock Exchange no. 871/2013 of August 6, 2013 as amended – concerning the WIG30 index. Indices Removal of the reference to NCIndex30.	Delivered
5.7	29/08/2016	FLO	-Order Book Retransmission Delimiter - 231 Message: clarification on sending message rules: A message is sent per instance of the trading engine (identified by InstanceNum).	Delivered
5.8	31/10/2016	TSE	- Trade Cancellation – 221 Message: Message removal -Trade Full Information – 240 Message: New fields are added to consider MIFID2 requirements -Trade Publication – 243 Message: New fields are added to consider MIFID2 requirements -Referential - 556 Message:	Delivered

Version	Date	Author	Description	Status
			Addition of the new field 'MaxValue'. Field description: Maximum order value allowed for orders on that instrument	
5.9	30/11/2016	TSE	Reviewed version for Mifid compliance validation	Delivered
5.9a	06/02/2017	SNM	Correction of the sizes of the following messages: -Trade Full Information – 240 Message -Trade Publication – 243 Message -Referential - 556 Message Update of the following content: -Changes of the 2.4.1 Sector Classification for the main market segments of WSE -Removal of the 2.4.2 Sector Classification on NewConnect	Delivered
5.9b	17/02/2017	PVE	Offsets alignment in messages: -Trade Full Information – 240 Message -Trade Publication – 243 Message -Referential - 556 Message	Delivered
6.0	21/04/2017	SNM	Update of TransactionIDCode, PriceNotation and Price description on Trade Full Information – 240 Message and Trade Publication – 243 Message Replace NomMktPrice by NotionalAmount field on Referential - 556 Message Addition of NotionalCurrency field on Referential - 556 Message	Delivered
6.1	12/06/2017	SNM	Update of TickDirection field description on Trade Full Information – 240 Message and Auction Summary – 245 Message Update of NotionalAmount, Price and Quantityfield description on Referential – 556 Message, Trade Full Information – 240 Message and Trade Full Information – 243 Message Replace NomMktPriceScaleCode, FixPriceTick and FixPriceTickScaleCode field by fillers on Referential - 556 Message	Delivered
6.2	23/08/2017	SNM	Update of TransactionToBeCleared field description on Trade Full Information – 240 Message and Trade Publication – 243 Message Update of offsets on Referential - 556 Message Update of description of MMT flags on Trade Full Information – 240 Message and Trade Publication – 243 Message	Delivered
6.3	10/10/2017	SNM	Update of Price, Quantity, Notional Amount, TransactionToBeCleared field description on Trade Full Information – 240 Message Update of InternalMsgType and respective body message on Generic WSE Message – 592 Message	Delivered
6.4	26/01/2018	SNM	Update of the following content: -Message description on Trade Full Information – 240 Message and Trade Publication – 243 Message - MMTTradingMode field on Trade Full Information – 240 Message -Section 2.8 MIC Codes: addition of WIPO and WOPO MIC.	Delivered
6.5	16/11/2018	KME	The change relates to 516, 547, 548 messages and apply to new strategy indices identified by: - message: 548, - field: 'ProductClassification', - value: 'M' - WSE strategy dividend point index Some fields will not be used (should be ignored) for the new index. Some fields have been updated with identification of new strategy index. Field 'NumberOfComponents' in 548 message will be field in with the number of dividends included in the dividend point index calculation. Additionally some editorial changes have been made.	
6.6	04/09/2019	GLE	Update the following content: - Session Summary – 561 message Field 'Quotation System' – new value added '81' "Single Price Quotation System with a single fixing" (incorporation of disclosed Extension no 1 of XDP v.6.4 dated 06.06.2018 - appendix 2.8 MIC Codes where: a) MIC WIPO – description change; b) MIC WOPO – decommissioned; c) MIC XOFF – added; d) MIC XXXX – added. MICs may appear in Referential – 556 Message as of 30 September 2019. Affected MICs used only for IPO and SPO procedures. - Appendix 2.8 MICCodes – removing values PLPX and WGAS	
6.7	30/10/2019	JMA	Message 241 Price Update Extension of the field "TypeOfPrice" with values:	

Version	Date	Author	Description	Status
			77 - Reference Price TBSP.Price 78 - Reference Price TBSP.fixPrice 88 - MidPrice Editorial changes: Description of value “34" changed: From: “New previous day’s closing price” To: “Reference Price” And inserted changed description (instead of the two akapits at the end of field’s description): For BondSpot Regulated Market and Alternative Trading System stream price is calculated in the following way: - Price of the last trade; - In case of lack of trade – issue price. For TBSP stream the price is calculated in the following way: - Last TBSP.fixPrice - In case of lack of TBSP.fixprice minimum auction price from the last auction.	
6.8	11/02/2020	MSM	- Update of TransactionIDCode field – clarification on message 243, - Update of StrikeCurrency field on message 556, - Update of TypeOfCorporateEvent field on message 556.	

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1 Real Time Message Specifications

1.1 Common Packet Header Format

All messages are preceded by a common packet header format. The following table describes the header fields of a Market Information message.

Field Name	Off set	Size (Bytes)	Format	Description
PacketLength	0	2	Binary Int.	Length of the packet including the 16-byte packet header.
PacketType	2	2	Binary Int.	Identifier for the type of data contained in the packet. Valid values: 501 – Market Data messages
PacketSeqNum	4	4	Binary Int.	This field contains the packet sequence number. It is unique for each broadcast stream (multicast group) and is used for gap detection. It increases serially and monotonically and is reset to 1 at the beginning of each trading day. The PackSeqNum is unique for packets containing market data only. Heartbeats inherit their sequence number from the last market data packet or packet sequence number reset packet.
SendTime	8	4	Binary Int.	Timestamp in millisecond indicating the packet broadcast time. The number represents the number of milliseconds since midnight of the last Sunday 00:00 UTC.
ServiceID	12	2	Binary Int.	Numeric value identifying the broadcast stream. Possible values are described in XDP CDE Configuration document
DeliveryFlag	14	1	Binary Int.	Indicates delivery method. Valid values: 0 Real Time message (Uncompressed) 2 Retransmission message (Uncompressed) 16 Real Time message (ZliB Compressed) 17 Refresh message (ZliB Compressed) 18 Retransmission message (Compressed)
NumberMsgEntries	15	1	Binary Int.	The number of messages that are contained within the packet.

1.2 MARKET INFORMATION

1.2.1 Overview

The Market Information service uses the push-based publishing model. This means that data will be published based on its availability. Once information is available, it will be published to Market Information clients. The Market Information message reflects the last information for each WSE traded instrument and information on the market itself.

List of the message types in the WSE Market Information Feed:

- 505 - Instrument state change message – 52 bytes in total, **50** excluding MsgSize field
- 516 - Class state change message – 28 bytes in total, **26** excluding MsgSize field
- 523 – Mail – 1102 bytes in total, **1100** excluding MsgSize field
- 530 - IMP (Indicative Matching Price) – 36 bytes in total, **34** excluding MsgSize field
- 537 – Collars – 36 bytes in total, **34** excluding MsgSize field
- 539 - Session Timetable – 144 bytes in total, **142** excluding MsgSize field
- 550 - Start referential – 8 bytes in total, **6** excluding MsgSize field
- 551 - End referential – 8 bytes in total, **6** excluding MsgSize field
- 556 – Referential – 544 bytes in total, **542** excluding MsgSize field
- 560 – Short Sale Change – 128 bytes in total, **126** excluding MsgSize field
- 561 – Session Summary – 264 bytes in total, **262** excluding MsgSize field
- 592 – Generic WSE message – 268 bytes in total, **266** excluding MsgSize field
- 595 – Open position – 40 bytes in total, **38** excluding MsgSize field

List of the message types in the BondSpot Market Information Feed:

- 516 - Class state change message – 28 bytes in total, **26** excluding MsgSize field
- 523 – Mail – 1102 bytes in total, **1100** excluding MsgSize field
- 537 – Collars – 36 bytes in total, **34** excluding MsgSize field
- 539 - Session Timetable – 144 bytes in total, **142** excluding MsgSize field
- 550 - Start referential – 8 bytes in total, **6** excluding MsgSize field
- 551 - End referential – 8 bytes in total, **6** excluding MsgSize field
- 556 – Referential – 524 bytes in total, **522** excluding MsgSize field
- 561 – Session Summary – 256 bytes in total, **254** excluding MsgSize field

1.2.2 Instrument State Change - 505 Message

1.2.2.1 Message Overview

An 'Instrument State Change' message is generated by the trading engines to announce a change in the state of an instrument.

1.2.2.2 Message Sending Rules

This message is sent for tradable instruments:

- When an instrument has been halted
- When an instrument has been unhalted
- When an unhalt (opening) is programmed
- When a programmed unhalt (opening) is cancelled
- Prior to the trading session, in order to carry over states of instruments halted the day before
- When a one side only period starts or ends for a warrant

1.2.2.3 Message Structure

The table below describes the body fields of an Instrument State Change message (MsgType = '505').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	505 – Instrument State Change
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
SourceTimeMicroSecs	20	2	Binary Int.	Number of micro seconds to be combined with SourceTime
Filler	*22*	*2*	*Binary*	*For future use.*
StartDateHalting	24	8	ASCII Str.	Date when suspension of trading started for an instrument. Valid values: YYYYMMDD 00000000 if not provided
StartTimeHalting	32	6	ASCII Str.	Time when suspension of trading started for an instrument. Valid values: HHMMSS 000000 if not provided
ProgOpeningTime	38	6	ASCII Str.	Instrument opening (unhalting) time that has been programmed by the Market Operation Team.Valid values: HHMMSS 000000 if not provided.
OrderEntryRejection	44	1	ASCII CH.	Indicates whether order entry is allowed or forbidden. Valid values: 'N' - Order entry allowed 'Y' - Order entry forbidden Null - Not provided
InstrumentState	45	1	ASCII Ch.	Indicates the state of the instrument. Valid values: 'H' - Halted Null (or space) Inherited (following the state of the class the instrument belongs to) 'A' - Auction Remark

Field Name	Offset	Size (Bytes)	Format	Description
				There is a relation between ClassState/Class (Msg 516) and InstrumentState: in case the state of the Class to which the instrument belongs differs from the state of that individual instrument, the most restrictive value supersedes: If the Instrument State shows “tradable” but its class is declared “not tradable” (halted), effectively trading is not allowed in this instrument.
InstrumentTradingStatus	46	1	ASCII Ch.	Not used
HaltReason	47	1	ASCII Ch.	Indicates the origin of halting for an instrument Valid values: ‘R’ - Halted. No liquidity provider 'C' - Trade price outside dynamic collars 'M' - Manual halting by Market Operations ‘W’ - Regulatory halting, deactivating barrier reached ‘X’ - Regulatory halting, underlying halted 'G' - Regulatory halting by MO 'S' - Trade price outside static collars 'A' - Number of Orders Alert Null or space - Instrument not halted or information not available
ActionAffectingState	48	1	ASCII Ch.	Code indicating the event that caused a change in the instrument state. Valid values:
				'C' - Trading on the instrument at the opening (sent before O) 'D' - Cancelled programmed opening 'M' - Instrument halted independently from its class ‘N’ - Instrument is being initialized (beginning of the trading day) 'O' - Instrument opened 'P' - Deferred programmed opening 'R' - Automatic halting at the class auction ‘Y’ - Beginning of a one side only period ‘Z’ - End of a one side only period
InstrumentStateTCS	49	1	ASCII Ch.	Not used
PeriodSide	50	1	ASCII Ch.	Indicates the side for an instrument beginning or ending a Bid only or Offer only situation. Valid values: '0' - Not applicable ‘1’ - Bid only ‘2’ - Offer only
Filler	*51*	*1*	*Binary Int.*	*For future use.*

1.2.2.4 Instrument State Change Parameters

The Instrument State Change Matrix describes the possible combination fields of an instrument state change :

Instrument State Change Matrix	XDP 505	HaltReason **(blank)** Inherited or info not available **C** Trade Price Outside Dynamic Collars **M** Manual halting by MO **W** Regulatory halting, deactivating barrier reached **X** Regulatory halting, underlying halted **R** Halted no LP **G** Regulatory Halted by MO **S** Trade Price Outside Static Collars **A** NSO Alert	InstrumentState **(blank)** Inherited or info not available **A** Auction **H** Halted	ActionAffectingState **(blank)** Not provided **P** Deferred programmed opening **M** Instrument halted independently from its class **C** Trading on the instrument at the opening **O** Instrument opened **R** Automatic halting at the class auction **D** Cancelled programmed opening **N** State at Initialization (Start of trading day) **Y** Start of one side only period **Z** End of one side only period	OrderEntryRejection **(blank)** Not provided **N** Order entry allowed **Y** Order entry forbidden	PeriodSide **(blank)** Normal Situation **1** Bid Only **2** Offer Only	UTM Instrument State
Halt cases							
Instrument standard halted by Market Operations in Core Continuous phase with order entry allowed		M	H	M	N	blank	Halted
Instrument regulatory halted in Core Continuous phase with order entry forbidden		G	H	M	Y	blank	Reg-Halted
Instrument halted by the system at the Core Auction due to the breaching of expanded dynamic collars		C	H	R	<current value>	blank	Halted
Instrument halted by the system due to the breaching of static collars at the Core Auction		S	H	R	<current value>	blank	Halted
Instrument halted by the system at the Core Auction due to the presence of MTL order with no orders on the opposite side of the order book		C	H	R	<current value>	blank	Halted
Instrument halted by the system in Core Continuous phase due to the breaching of static collars		S	H	M	<current value>	blank	Halted
Instrument technical halted by the system in Core Continuous phase		blank	H	blank	Y	blank	Tech-Halted
Trading Engine Initialization (if Symbstatus = H and/or OrderEntryRejection = Y)		blank	blank H H	N	Y N Y	blank	Tech-Halted
Instrument halted by the system due to the absence of MM orders at the Core Auction		R	H	R	<current value>	blank	Waiting for LP
Instrument halted by the system due to the absence of MM orders in Core Continuous phase		R	H	M	<current value>	blank	Waiting for LP
Instrument halted by the system due to the virtual offer breaching		C	H	M	<current value>	blank	Halted
Instrument halted by the system due to the reaching of deactivating barrier		W	H	M	<current value>	blank	Halted
Warrant Instrument halted due to the halt of its underlying		X	H	M	<current value>	blank	Halted
Instrument halted by the system due to the breaching an NSO alert (max number of orders on an instrument)		A	H	M	Y	blank	Halted
Unhalt cases							
Instrument unhalted by Market Operations in Call phase		blank	blank	blank	<current value>	blank	Inherited
Instrument unhalted by Market Operations in Core Continuous phase with matching at the opening (i.e. at least one trade has been generated) (two XDP 505 are generated)	first 505	blank	H	C	<current value>	blank	Inherited
	second 505	blank	blank	O	<current value>	blank	Inherited
Instrument unhalted by Market Operations in Core Continuous phase without matching at the opening		blank	blank	O	<current value>	blank	Inherited
Sent LP Quotes on a "waiting for LP" instrument		blank	blank	O	<current value>	blank	Inherited
Auction by Market Operations with trade generation with matching at the opening (i.e. at least one trade has been generated) (two XDP 505 are generated)	first 505	blank	A	C	<current value>	blank	Inherited
	second 505	blank	blank	O	<current value>	blank	Inherited
Auction by Market Operations without matching at the opening		blank	blank	O	<current value>	blank	Inherited
Instrument unhalted during Class Auction without matching at the opening		blank	blank	O	<current value>	blank	Inherited
Instrument unhalted during Class Auction with matching at the opening (i.e. at least one trade has been generated) (two XDP 505 are generated)	first 505	blank	A	C	<current value>	blank	Inherited
	second 505	blank	blank	O	<current value>	blank	Inherited
Other cases							
Trading Engine Initialization (if Symbstatus = H or OrderEntryRejection = Y)		blank	<current value>	N	<current value>	blank	Tech-Halted
Deferred Opening		blank	<current value>	P	<current value>	blank	Current State
Cancelled Deferred Opening		blank	<current value>	D	<current value>	blank	Current State
Instrument begins a Bid Only period		blank	<current value>	Y	<current value>	1	Current State
Instrument begins an Offer Only period		blank	<current value>	Y	<current value>	2	Current State
Instrument ends a Bid Only period		blank	<current value>	Z	<current value>	1	Current State
Instrument ends an Offer Only period		blank	<current value>	Z	<current value>	2	Current State

1.2.3 Class State Change - 516 Message

1.2.3.1 Message Overview

The Class State Change message indicates a change in the state of a class.

There is a relationship between ClassState (516 message) and InstrumentState (505 message). In case the state of the Class to which the instrument belongs differs from the state of that individual instrument, the most restrictive value supersedes the other. If the Instrument State shows “tradable” but its class is declared “not tradable” (e.g. halted), effectively trading is not allowed in this instrument.

1.2.3.2 Message Sending Rules

This message is sent each time a class state changes during the trading day.

1.2.3.3 Message Structure

The table below describes the body fields of a Class State Change message (MsgType = ‘516’).

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	516 – Class State Change
SourceSeqNum	4	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SystemID	12	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	16	2	Binary Int.	Number of micro seconds. To be combined with SourceTime. Always filled with 0 for BondSpot.
InstrumentGroupCode	18	3	ASCII Str.	Class Identification
InstrumentGroupState	*21*	1	ASCII Ch.	Not used
SessionType	22	1	ASCII Ch.	Market Session. Valid values: ‘E’ - Early (not used) ‘C’ - Core Session ‘L’ - Late (not used)
Filler	23	*1*	*Binary*	*For future use.*

Field Name	Offset	Size (Bytes)	Format	Description
ClassState	24	4	ASCII CH.	Class status representing the current trading market phase for instruments belonging to that class and whose status is inherited. Valid values: 'EAMO' - Early Monitoring 'COCA' - Core Call 'COAU' - Core Auction 'COCO' - Core Continuous 'CLCA' - Closing Call 'CLAU' - Closing Auction 'LTAL' - Late Trading At Last 'CTAL' - Core Trading At Last 'COMO' - Core Monitoring 'LAMO' - Late Monitoring 'HALT' - Halted 'CLSD' - Closed

1.2.4 Mail - 523 Message

1.2.4.1 Message Overview

This message supplies a part or a complete electronic message (E-mail), in order to inform members and rebroadcasting members about quotation conditions on a given listed instrument or a set of listed instruments or otherwise about a technical problem.

1.2.4.2 Message Sending Rules

This message is sent manually by Market Operation Team to inform member firms about events of general interest that occurred in the market (suspension of securities, deletions of order books, new listings of securities, various technical messages...).

1.2.4.3 Message Structure

The table below describes the body fields of a Mail message (MsgType = '523').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	523 - Mail
SourceSeqNum	4	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SystemID	12	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	16	2	Binary Int.	Number of micro seconds. To be combined with SourceTime. Always filled with 0 for BondSpot.
Scope	18	1	ASCII Ch.	Scope. Valid values: '1' - Warsaw Stock Exchange Markets

Field Name	Offset	Size (Bytes)	Format	Description
Title	*19*	80	ASCII Ch.	Title.
Text	99	1000	ASCII Str.	Text.
Filler	*1099*	*3*	*Binary*	*For future use*

1.2.5 Indicative Matching Price (IMP) - 530 Message

1.2.5.1 Message Overview

The Indicative matching price message indicates an instrument's theoretical opening price (also known as an indicative opening price):

- An IMP-message is sent if the indicative matching price or – volume varies.
- If the indicative matching price remains undetermined, but the reason for this indetermination changes (for example following a cancellation or modification of an order), then a Theoretical opening price message is sent with zero values.

1.2.5.2 Message Sending Rules

- When an instrument is in call mode, this message broadcasts the Indicative Matching Price (IMP) every time this price changes in asynchronous mode. On the UTP trading engine, the IMP is calculated in a preset frequency (set at 50 msec for WSE);
- At the beginning of an auction processing on an instrument (before the first trade), it provides the auction price (that may differ from the previous IMP);

1.2.5.3 Message Structure

The table below describes the body fields of an Indicative Matching Price message (MsgType = '530').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	530 - Indicative Matching Price
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
IMPrice	16	4	Binary Int.	Indicative Matching Price: Price at which transaction on a security would occur if Opening occurred at this moment (To be calculated with the PriceScaleCode).

Field Name	Offset	Size (Bytes)	Format	Description
Variation	20	4	Binary Int. (signed)	Percentage variation of the last IMP against the previous day's reference price for the stock concerned, ie the last known price can be the last adjusted closing price or the last price indication (valuation trade).(To be calculated with the VariationScaleCode).
SystemID	*24*	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	28	2	Binary Int.	Number of micro seconds. To be combined with SourceTime.
PriceScaleCode	30	1	Binary Int.	To be combined with IMPrice
VariationScaleCode	31	1	Binary Int.	To be combined with Variation
IMVolume	32	4	Binary Int.	Indicative Matching Volume: Volume that would be exchanged if Auction occurred at this moment.

1.2.6 Collars - 537 Message

1.2.6.1 Message Overview

A Collars message informs Clients of modifications in authorized price fluctuations for an instrument.

1.2.6.2 Message Sending Rules

This message is sent:

- At the start-up of the UTP trading system in the morning
- Each time Cash Market Operations changes a dynamic collar / static threshold for an instrument
- After the first auction of the given instrument.
- After each change of the value of collars

1.2.6.3 Message Structure

The table below describes the body fields of a Collars message (MsgType = '537').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	537 - Collars
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
HighCollar	16	4	Binary Int.	Higher bound of the collar price range (To be calculated with the HighScaleCode).

Field Name	Offset	Size (Bytes)	Format	Description
LowCollar	20	4	Binary Int.	Lower bound of the collar price range (To be calculated with the LowScaleCode).
SystemID	*24*	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
SourceTimeMicroSecs	28	2	Binary Int.	Number of micro seconds. To be combined with SourceTime.
HighScaleCode	30	1	Binary Int.	To be combined with HighCollar
LowScaleCode	31	1	Binary Int.	To be combined with LowCollar
CollarType	32	1	ASCII Ch.	Defines the type of collars. Valid values: 'D' - Dynamic collars 'S' - Static thresholds
Filler	*33*	*3*	Binary NULL	*For future use*

1.2.7 Session Timetable - 539 Message

1.2.7.1 Message Overview

The session timetable message indicates the timetable listing all the states changes of a Class for the current trading day.

1.2.7.2 Message Sending Rules

This message is sent:

- Automatically for each Class at the beginning of the trading day, to indicate the times at which the market session will change from one phase to another
- On an exceptional basis, it may be sent during the trading day in cases where normal hours have changed during the day.

1.2.7.3 Message Structure

The table below describes the body fields of a Session timetable message (MsgType = '539').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	539- Session timetable
SourceSeqNum	4	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SystemID	12	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
SourceTimeMicroSecs	16	2	Binary Int.	Number of micro seconds. To be combined with SourceTime. Always filled with 0 for BondSpot.

Field Name	Offset	Size (Bytes)	Format	Description
InstrumentGroupCode	18	3	ASCII Str	Indicates on which Class the session timetable is applied
Filler	*21*	*1*	*Binary*	*For future use*
ClassPhaseNumber	22	2	Binary Int.	Indicates the number of class phases
ClassStateTime	24 (mod 8)	4	ASCII Str.	Indicates the start time of the class phase, format HHMM. 0000 phase linked to the previous phase. Repeated 15 times together with ClassState. Time use CET.
ClassState	28 (mod 8)	4	ASCII CH.	Class Status representing the current trading market phase. Repeated 15 times together with ClassStateTime. Valid values: 'EAMO' - Early Monitoring 'COCA' - Core Call 'COAU' - Core Auction 'COCO' - Core Continuous 'CLCA' - Closing Call 'CLAU' - Closing Auction 'LTAL' - Late Trading At Last 'CTAL' - Core Trading At Last 'COMO' - Core Monitoring 'LAMO' - Late Monitoring 'CLSD' - Closed Blank (spaces) - Inactive class phase

1.2.8 Start Referential – Message 550

1.2.8.1 Message Overview

The Start referential message is sent to signal the start of transmission of the Instrument Reference Data Flow. It is sent in the morning and in the evening.

1.2.8.2 Message Sending Rules

This message is sent:

- Each time referential messages (556) are sent, prior to the first one.

1.2.8.3 Message Structure

The table below describes the body fields of a Start Referential message (MsgType = '550').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	550 - Start referential
Indicator	4	1	ASCII Ch.	This field indicates the start of the Referential flow. Always takes the value 'S'
Filler	*5*	*3*	*Binary*	*This is a filler, reserved for a future use*

1.2.9 End Referential - 551 Message

1.2.9.1 Message Overview

The End referential message is sent to signal the end of transmission of the Instrument Reference Data Flow. It is sent in the morning and in the evening.

1.2.9.2 Message Sending Rules

This message is sent:

- Each time referential messages (556) are sent, after the last one.

1.2.9.3 Message Structure

The table below describes the body fields of an End Referential message (MsgType = '551').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	551- End referential
Indicator	4	1	ASCII Ch.	This field indicates the end of the Referential flow. Always takes the value 'E'
Filler	*5*	*3*	*Binary*	*This is a filler, reserved for a future use*

1.2.10 Referential - 556 Message

1.2.10.1 Message Overview

The referential message indicates the main characteristics of a listed instrument:

- Characteristics of the instrument itself
- Trading characteristics of the instrument
- Previous trading day price.

For morning messages, the characteristics are valid for the trading day at the start of which it was transmitted.

For evening messages, the characteristics are valid for the next business date (the trading day D+1 following the Day D of its transmission).

1.2.10.2 Message Sending Rules

This message is sent:

- Every morning. There is a 556 message per instrument, tradable or broadcasted for the current day.
- Every evening. There is a 556 message per instrument, tradable or broadcasted for the next trading day.
- 556 messages will be sent in between a start and an end reference data message (550 and 551).

1.2.10.3 Message Structure

The table below describes the body fields of a Referential message (MsgType = '556').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	556 - Referential
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
LastAdjPrice	16	4	Binary Int.	Last traded price of the previous trading day after application of the adjustment coefficient (To be calculated with the LastAdjPriceScaleCode). For BondSpot: The field could be filled with zeros. In case of receiving the message where the field LastAdjPrice is filled with zeros then the value should not be published. For BondSpot the Last Adjusted Price is calculated in the following way:
				1. Price of the last trade 2. In case of lack of trade – issue price or nominal value for securitites For BondSpot TBS Poland stream the LastAdjPrice is calculated in the following way: 1. Price of the last fixing 2. In case of lack of price of the last fixing – price of the last trade 3. In case of lack of the last trade volume-weighted average price from the latest primary market Auction.
SystemID	20	4	Binary Int.	The ID of the originating Exchange/System of the message
PrevVolumeTraded	24	4	Binary Int.	Not used
Filler	28	4	*Binary*	*For future use*
SourceTimeMicroSecs	32	2	Binary Int.	Number of micro seconds. To be combined with SourceTime
Stock Exchange Code	34	2	Binary Int.	Indicates the Market Place. Valid values: Refer to appendix Stock Exchange Code
TypeOfInstrument	36	2	Binary Int.	Type of instrument. Valid values: Refer to appendix Stock Type.
EventDate	38	8	ASCII Str.	Date of the last modification of the characteristics of the instrument except the following ones (which are modified every day):

Field Name	Offset	Size (Bytes)	Format	Description
				- Previous day's adjusted closing price (LastAdjPrice) - Previous day capital traded (PrevDayCapitalTraded) - (Not used) - Number of shares for this security traded on previous day (PrevVolumeTraded)) - (Not used) - Date when the security last traded (DateOfLastTrade) - (Not used) Format Date: YYYYMMDD [
InstrumentName	46	18	ASCII Str.	Instrument Name For instrument others than SHARES: Contains a short instrument name (Formally known as mnemonic)
PeriodIndicator	64	1	ASCII Ch.	Not used
TypeOfMarketAdmission	65	1	ASCII Ch.	Not used.
IssuingCountryCode	66	3	ASCII Str.	Country code of location for the corporate headquarters of the company that issued the instrument. (ISO 3166-3A)
TradingCurrency	69	3	ASCII Str.	Code of the currency in which the instrument is traded. (ISO 4217-3A)
InstrumentGroupCode	72	3	ASCII Str.	Designates the class to which the security belongs.
InstrumentCategory	75	1	ASCII Ch.	Indicates to which category, the security belongs. Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'A' - Stock 'O' - Bond 'W' - Warrant and certificate 'T' - Tracker 'I' - Index 'F' - Future 'X' - Option Additional possible values for BondSpot: 'Z' - Bond quoted on information fixing 'O' - Treasury bills and bonds not quoted on information fixing
InstrumentCode	76	12	ASCII Str.	ISIN Code, International Securities Identification Number, according to the ISO-6166.
DateOfLastTrade	88	8	ASCII Str.	Not Available
UnderlyingRepoISINCode	96	12	ASCII Str.	Underlying security (security used in the loan quotation system) for loan contracts at given value on centralized lending market. Used on L&B segment only.(Lending and Borrowing market segment) Not used.
ExpiryDate	108	8	ASCII Str.	Instrument expiry date. (YYYYMMDD)

Field Name	Offset	Size (Bytes)	Format	Description
FirstSettlementDate	116	8	ASCII Str.	Represents the first possible settlement date for a given instrument with the instrument's depository. When this date is not provided, it means that the first possible settlement date is the same as the instrument's flotation date. Not used.
TypeOfDerivatives	124	1	ASCII Ch.	Type of derivative instrument associated to the security. Valid values: '1' - INAV (Indicative Net Asset value for ETF-Exchange Traded Funds) '4' - Supports short terms options, '6' - Supports long term options '9' - No
BICDepositary	125	11	ASCII Str.	Identifies the depository organization for the shares of an instrument, or when the same organization manages several systems, this item identifies the relevant settlement system for settling trades on a given listed security. The ID is allocated by SWIFT, according to the BIC standard (ISO 9362). Not used.
ICB	136	4	ASCII Str.	Identifies for a listed instrument, the economic subsector of the issuing company in the ICB (Industry Classification Benchmark) classification.

Field Name	Offset	Size (Bytes)	Format	Description
MIC	140	4	ASCII Str.	Identifies the market to which an instrument belongs to by its MIC (Market Identification code), according to ISO 10383. Valid values: Refer to appendix MIC Codes
UnderlyingISINCode	144	12	ASCII Str.	Gives the ISIN Code of a warrant or a derivatives instrument.
DepositaryList	156	25	ASCII Str.	Identifies the possible main depository organizations (max 5 characters padded with zeros) for the shares or bonds for an instrument. Used by the clearing house to determine the relevant system for settling trades. Valid values: '00001' - KDPW '00000' - No depository organization Null - Not significant
MainDepositary	181	5	ASCII Str.	Identifies the default (or main) depository organisation of the instrument used by priority for the settlement (e.g.: multi-listed instruments which have several depositories). This data has to be treated in consideration of the data DepositoryList. Used by the clearing house to determine the relevant system for settling trades. Valid values: Same as data "DepositoryList"

Field Name	Offset	Size (Bytes)	Format	Description
TypeOfCorporateEvent	186	2	ASCII Str.	Indicates the last type of corporate event that has occurred on an instrument, such as detachment of rights, or of coupons. The data item is automatically calculated by the adjustment application but in case of problem or error, the data item value could be modified manually, particularly for purging the order book in case of absence of corporate event. This data has to be treated in consideration of the date of the event included into the header of the message. Valid values: '00' - No specific event '01' - Dividend payment in cash or in stocks '03' - Interest payment (bonds for which the price is expressed in % of the nominal) '04' - Split '05' - Bonus (i.e. attribution) '09' - Reverse split '12' - Purge of the order book in the absence of a corporate event, e.g. following the modification of the variable tick of the listed instrument '17' - Spin-off '18' - Rights '19' - Bonus and Rights '20' - Bonus also entitled for Rights '21' - Rights also entitled for Bonus

Field Name	Offset	Size (Bytes)	Format	Description
				'99' - Multiple Corporate Events
TimeLagEuronextUTC	188	5	ASCII Str.	Not used.
TimeLagMiFIDRegUTC	193	5	ASCII Str.	Not used.
CFI	198	6	ASCII Str.	Classification code of a financial instrument defined by the ISO-10962 standard. The structure of the CFI code: - The CFI reflects characteristics that are defined when a financial instrument is issued, and remain unchanged during its entire lifetime. - The CFI consists of six alphabetical characters: 1st character indicates the highest level of classification (Categories) 'E' - Equities 'D' - Debt instruments 'R' - Entitlements (Rights) 'O' - Options 'F' - Futures 'M' - Others/Miscellaneous 2nd character indicates specific groups within each category: Groups e.g. for equities: - Shares - Preferred shares - Convertible preferred shares - Units, i.e. unit trusts/mutual funds etc. - Others

Field Name	Offset	Size (Bytes)	Format	Description
				3rd to 6th character indicate the most important attributes to each group: Attributes e.g. for equities: - Voting right - Ownership/transfer restrictions - Payment status - Form
Filler	204	4	*Binary*	*For future use*
IndexSetOfVarPriceTick	208	2	ASCII Str.	When the instrument's tick is defined via a Variable Tick Table: The key for the variable price tick table consists of two data items [the index for a set of variable price ticks, the lowest value in a range of prices]. The index data item refers to a set of lines in that table which make it possible to determine the price tick for an instrument, based on the price range in which a given price for the instrument falls (i.e. a price to be rounded off or a limit to be checked).
MarketFeedCode	210	2	ASCII Str.	Not Available
MICList	212	24	ASCII Str.	Not used.
IndustryCode	236	4	ASCII Str.	Not used.
IndustryText	240	100	ASCII Str.	Not used.
FinancialMarketCode	340	3	ASCII Str.	Not Used
USIndicators	343	7	ASCII Str.	Not used.
Filler	350	*2*	*Binary*	*For future use*
PrevDayCapitalTraded	352	8	Binary Int.	Not used.

Field Name	Offset	Size (Bytes)	Format	Description
LotSize	360	8	Binary Int.	Expressed in number of shares or in an amount or a volume of the capital, of the lot size. The lot size is a minimum tradable quantity that is set for each instrument. The quantity of an order entered by a trading member on the market must be a multiple of the lot size. For bonds, this data has to be considered with the data "Amount of par value for instrument for calculating trade amount". This item is calculated in the following way: Note 1 Only positive values are possible. If not provided, will be set to 0 Note 2 Only integer values that are equal to or greater to one are accepted. According to LotSizeScaleCode.
NumberInstrumentCirc	368	8	Binary Int.	Number of shares issued/bonds outstanding after payments
SharesOut	376	8	Binary Int.	For BondSpot ONLY: Number of quoted instrument For WSE: filled with zero.
AuthShares	384	8	Binary Int.	Not used.
PriceNotation (new field in place of Filler offset 400 (3) and TypeOfUnitExp from offset 405 (1) that is removed)	392	4	ASCII Str.	Indication as to whether the price and the strike price is expressed in monetary value, in percentage or in yield Possible values: - MONE: monetary value. - PERC: percentage. - YIEL: yield. - BAPO: basis points.

Field Name	Offset	Size (Bytes)	Format	Description
As a consequence only RepoIndicator and LastAdjPriceScaleCode changed their offsets				
RepoIndicator	396	1	ASCII Ch.	Not used.
NotionalAmount (new field)	397	20	ASCII Ch.	Nominal amount or notional amount. Possible values: – maximum of 18 digits with a maximum of 5 decimals Where applicable the following format should be considered: note 1: decimal separator is '.' (full stop) note 2: negative numbers are prefixed with '-' (minus) note 3: where applicable, values shall be rounded and not truncated note 4: the value will be padded on the left side with '0' (zero)
NotionalCurrency (new field)	417	3	ASCII Ch.	Currency in which the notional is denominated as defined by ISO 4217.
LastAdjPriceScaleCode	420	1	Binary Int.	To be combined with LastAdjPrice.
MarketIndicator	421	1	Binary Int.	Indicates the market regulations governing the market on which the security is traded. Valid values: 0 Not used
PrevDayCapitalTradedScaleCode	422	1	Binary Int.	Not used.
TaxCode	423	1	Binary Int.	Not used.
Filler	424	1	*Binary*	*For future use*
LotSizeScaleCode	425	1	Binary Int.	To be combined with LotSize.
Filler	426	1	*Binary*	*For future use*
Mnemo	427	5	ASCII Str.	Mnemonic code of the instrument. This is to be used only for Shares.
TradingCode	432	12	ASCII Str.	For WSE: Trading code of the instrument. For BondSpot: Technical code, not to be used. Please ignore it.
Filler	444	*4*	*Binary*	*For future use*
StrikePrice	448	4	Binary Int.	The specified price of an option contract at which the contract may be exercised, whereby a call option buyer can buy the underlying or a put option buyer can sell the underlying. The buyer's profit from exercising the option is the amount by which the strike price exceeds the cash instrument price (in the case of a call), or the amount by which the cash instrument price exceeds the strike price (in the case of a put). In general, the smaller the difference between spot (cash instrument price) and strike price, the higher the option premium. Also called exercise price. According to StrikeScaleCode. Valid values: Only provided for warrants or other derivative instruments.
StrikeCurrency	452	3	ASCII Str	This field contains the currency in which the Warrant Strike Price is expressed Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				Only provided for warrants or other derivative instruments.
StrikeScaleCode	455	1	Binary Int.	To be combined with StrikePrice
CurrencyCoef	456	4	Binary Int.	Change ratio coefficient of currency applied to instrument: Used in conjunction with one of the change rate indicators in order to apply this coefficient to a currency among two available currencies defined for the instrument: - Trading currency - Currency code of strike price for Derivative Instrument The currency to which this coefficient will be applied depends on just one of the two values set to the related indicators (defined below in the integrity constraints). This coefficient is used when the currency is not compliant with the ISO 4217 (3A) standard as pence (GBP) or cent (USD) expression of an official currency. In this case the formula to apply in order to retrieve the price expressed in an official currency is: Real price in Trading currency = Traded price (i.e. 1565 pence) x Change ratio coefficient value (0.01) Not used.
CurrencyCoefScaleCode	460	1	Binary Int.	To be combined with CurrencyCoef Not used.
TradingCurrencyIndicator	461	1	Binary Int.	Change rate indicator for currency of Instrument Traded Price:
				Indicates if the change rate will be applied to the currency defined for traded prices of the instrument. Valid values: Null Not applicable
StrikeCurrencyIndicator	462	1	Binary Int.	Change rate indicator for currency of the strike price Indicates if the change rate will be applied to the currency defined for strike prices of the instrument. Valid values: 0 Change rate not applied to the strike price. 1 Change rate applied to the strike price. Null Not applicable
MarketSegment	463	2	ASCII Ch.	Market Segment. Valid values: 'NM' - Cash Normal Market 'OT' - Cash Block Trading 'TO' - Tender Offer 'IP' - IPO 'IN' - Investment 'LE' - Leverage 'DN' - Derivatives Normal Market 'DB' - Derivatives Block Trading 'IX' - WSE Indices For BondSpot only:

Field Name	Offset	Size (Bytes)	Format	Description
				'BR' - Retail Market 'BW' - Wholesale Market 'BT' - BondSpot TBS Poland 'BA' - BondSpot ATS
Algo	465	1	ASCII Ch.	Indicates which algorithm is used. Valid values: 'F' - FIFO 'O' - FIFO Origin
LocalName	466	30	ASCII Ch.	Issuing company short name
BlockSize	496	4	Binary Int.	Size of a normal block
SSMarketMaker	500	1	ASCII Ch.	**Field should be ignored** Short sale eligibility for a capacity of a market maker. Valid values: 'Y' - Short sale is authorized 'N' - Short sale is forbidden Blank - Not significant for Warrants and Derivatives instruments, as well as instruments traded on IPO and TO segments.
SSNonMarketMaker	501	1	ASCII Ch.	**Field should be ignored** Short sale eligibility for a capacity different from market maker. Valid values: 'Y' - Short sale is authorized 'N' - Short sale is forbidden Blank - Not significant for Warrants and Derivatives instruments, as well as instruments traded on IPO and TO segments.

Field Name	Offset	Size (Bytes)	Format	Description
SettlementDelay	502	2	ASCII Ch.	Settlement delay for the trades.
OptionType	504	1	ASCII Ch.	Indicates the type of an option only (otherwise left blank i.e. space). Valid values: 'C' - Call 'P' - Put
DeliveryType	505	1	ASCII Ch.	Indicates the type of delivery for a derivatives instrument only (otherwise left blank i.e. space) Valid values: 'C' - Cash settlement 'P' - Physical delivery
ExerciseType	506	1	ASCII Ch.	Indicates the type of exercise for an option only (otherwise left blank i.e. space). Valid values: 'A' - American 'E' - European
MultiplierScaleCode	507	1	Binary Int.	To be combined with Multiplier.
WarrantThresholdMinScale Code	508	1	Binary Int.	To be combined with WarrantThresholdMin
WarrantThresholdMaxScale Code	509	1	Binary Int.	To be combined with WarrantThresholdMax
Filler	510	*2*	*Binary*	*For future use*
WarrantThresholdMin	512	4	Binary Int.	Minimum threshold for the warrant underlying. For Warrant ONLY, otherwise left blank (binary null) for all other types of instrument.
WarrantThresholdMax	516	4	Binary Int.	Maximum threshold for the warrant underlying.

Field Name	Offset	Size (Bytes)	Format	Description
				For Warrant ONLY, otherwise left blank (binary null) for all other types of instrument.
MinQuantity	520	4	Binary Int.	Minimum quantity allowed for orders on that instrument.
MaxQuantity	524	4	Binary Int.	Maximum quantity allowed for orders on that instrument.
Multiplier	528	4	Binary Int.	Multiplier.
Filler	532	*4*	*Binary*	*For future use*
MaxValue (new field)	536	8	Binary Int.	Maximum order value allowed for orders on that instrument.

1.2.11 Short Sale Change - 560 Message

Note

This message should be ignored.

1.2.11.1 Message Overview

The Short Sale Change message indicates whether an instrument short sale is suspended or re-authorized.

1.2.11.2 Message Sending Rules

This message is sent each time an instrument short sale eligibility is suspended or re-authorized.

1.2.11.3 Message Structure

The table below describes the body fields of a Short Sale Change message (MsgType = '560').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	560 – Short Sale Change
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
SourceTimeMicroSecs	20	2	Binary Int.	Number of micro seconds to be combined with SourceTime
Text	22	100	ASCII Ch.	Free text
Authority	122	1	ASCII Ch.	Indicates the authority which required the short sale change. Valid values: '1' - WSE

Field Name	Offset	Size (Bytes)	Format	Description
				'2' - KDPW '3' - KNF
SSMarketMaker	123	1	ASCII Ch.	Short sale eligibility for a capacity of a market maker. Valid values: 'Y' - Short sale is authorized 'N' - Short sale is suspended
SSNonMarketMaker	124	1	ASCII Ch.	Short sale eligibility for a capacity different from market maker. Valid values: 'Y' - Short sale is authorized 'N' - Short sale is suspended
Filler	*125*	*3*	*Binary*	*For future use*

1.2.12 Session Summary - 561 Message

1.2.12.1 Message Overview

This message contains statistical data. One message 561 is generated for each instrument. The message contains quotations of all instruments in continuous and single price quotations.

Note

This message is generated only for instruments from normal market segment. The fields regarding Block Trading segment (i.e. BlockTradesNumber, VolumeBlockTrades, ValueBlockTrades, VWAPBlockTrade, MaximumBlockPrice, MinimumBlockPrice) will include the proper values dedicated to block trades.

1.2.12.2 Message Sending Rules

This message is sent:

- After the end of the trading sessions.

1.2.12.3 Message Structure

The table below describes the body fields of a Session Summary message (MsgType = '561').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	561 – Session Summary
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit.

Field Name	Offset	Size (Bytes)	Format	Description
				In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
Date	16	4	Binary Int.	Stock Exchange date (YYYYMMDD).
Multiplier	20	4	Binary Int.	Instrument multiplier.
AccumulatedInterest	24	4	Binary Int.	Accumulated interest on the bonds or for mortgage-backed bonds on the day of settling the transaction.
OpeningPrice	28	4	Binary Int.	Single-price quotation: The first single-price defined at the fixing during a particular stock exchange session. In the absence of a single-price, the field is completed with zeros. Continuous trading: Opening price for continuous trading. In the absence of transaction at the opening, the price of the first transaction from a particular session. In the absence of transaction during a particular session the field is completed with zeros.
MaximumPrice	32	4	Binary Int.	Single-price quotation: The maximum single-price during the current stock exchange session. In the absence of transaction.during a particular session, the field is completed with zeros. Continuous trading:
				The maximum price during the current session. In the absence of transaction.during a particular session, the field is completed with zeros.
MinimumPrice	36	4	Binary Int.	Single-price quotation: The minimum single-price during the current session. In the absence of transaction during a particular session, the field is completed with zeros. Continuous trading: The minimum price during the current session. In the absence of transaction during a particular session, the field is completed with zeros.
ClosingPrice	40	4	Binary Int.	Single-price quotation: Single-price determined at the last fixing during a particular session. Continuous trading: Closing price for continuous trading. In the absence of transaction at the closing the price of the last transaction during a particular stock exchange session

Field Name	Offset	Size (Bytes)	Format	Description
				In the absence of transactions, the closing price is equal to the reference price for a particular session, or the last known LP midpoint (only for instruments from IN and LE market segments).
ImpliedVolatility	44	4	Binary Int.	Implied volatility of WIG 20 index calculated on the basis of the price of the last transaction at the session for options on WIG 20 index with the use of Black-Scholes option pricing model according to algorithm worked out by the WSE, which is available on the website: www.opcje.gpw.pl. Implied volatility is the basis for calculating volatility used to determine Greek indicators (delta, gamma, rho, theta, vega). The field is filled only for options for which there was trade at the session. For series of options for which there was no trade at the session and for other instruments, the field is filled with zeros.
AverageWeightedPrice	48	4	Binary Int.	The average weighted by the trading volume price of concluding the transaction for a particular instrument during a particular stock exchange session. This value is calculated based on the following algorithm :
				Σ(transaction price * transaction volume) / Σ (transaction volume) In the absence of transaction at a particular session, the field is completed with zeros.
ReferencePrice	52	4	Binary Int.	The reference price valid for the current session.
PctChangePrice	56	4	Binary Int.	Percentage change of the instrument closing price at the current session in relation to the reference price. For futures, the change of the settlement price from the current session in relation to the reference price.
DividendRate	60	4	Binary Int.	Dividend rate for the company is given on the basis of dividends paid by companies being the base instrument for futures and options. Dividend rate for the index is given on the basis of dividends paid by companies comprised in the index being the base instrument for options or futures. Dividend rate is determined for the period until the expiry of options or futures. Dividend rate is used to determine Greek indicators (delta, gamma, rho, theta, vega).

Field Name	Offset	Size (Bytes)	Format	Description
				Dividend rate is determined for futures and options whose base instrument are stocks or stock exchange indexes. For other shares, the field is filled with zeros.
SettlementPrice	64	4	Binary Int.	The price defined according to the trading conditions depending on the type of instrument. The field is filled in for the following instruments: • Index units • Futures contracts • Options • Structured certificates For other instruments, the field is completed with zeros.
SettlementValue	68	4	Binary Int.	The price defined depending on the type of instrument. • Index units: Settlement value = Settlement Price • Futures contracts: Settlement value = Settlement Price * multiplier • Bonds and instruments quoted in percentage of the nominal value: Settlement value = Closing price / Last fixing price * nominal value/100 + accumulated interest. (Until the first trade : reference price × nominal value /100 + accumulated interest) • Options: Settlement value = Settlement Price * multiplier • IPO: Settlement value = Price of the trades of IPO • Warrants: Settlement value = Closing Price • Instruments from NewConnect Market (price driven market): Settlement value = Price of the last trade. In case of the lack of the trade the field is filled in with zeros. • Structured certificates: Settlement value = The value calculated by Issuer on the basis of formula defined in Final Terms • Structured Bonds: Settlement value = The value calculated by Issuer on the basis of formula defined in Final Terms • Other instruments: Settlement value = Closing Price / Last Fixing Price In the absence of transaction at a particular session, the field is completed with zeros.
Filler	72	4	*Binary Int.*	For future use.
OpenPositions	76	4	Binary Int.	Number of open positions after the end of the session.

Field Name	Offset	Size (Bytes)	Format	Description
				This field is completed for the index units, futures, options. For other instruments, the field completed with zeros.
TradingVolume	80	4	Binary Int.	Accumulated trading volume of all transactions concluded on a particular instrument during the current session.
TransactionsNumber	84	4	Binary Int.	The total number of transactions concluded on a particular instrument during the current market session.
TradingValue	88	8	Binary Int.	Accumulated value of all transactions concluded on a particular instrument during the current stock exchange session. For instruments traded in currency other than PLN, value is converted on PLN according to currency price.
CurrencyPrice	96	4	Binary Int.	The field contains the currency price from the Stock Exchange trading session day. Currency price is provided with fixed number of six decimal places. Currency price is provided in PLN per currency unit. Example: National Bank of Poland publishes currency price for HUF per 100 units 100 HUF = 1,6378 PLN The value of fields in message is equal 0,016378 For instruments traded in PLN the value of field is equal 1.

Field Name	Offset	Size (Bytes)	Format	Description
Delta	100	4	Binary Int.	Value of Delta indicator from the current session. The indicator is calculated for options only. For other instruments, the field is filled with zeros.
Gamma	104	4	Binary Int.	Value of gamma indicator from the current session. The indicator is calculated for options only. For other instruments, the field is filled with zeros.
Rho	108	4	Binary Int.	Value of rho indicator from the current session. The indicator is calculated for options only. For other instruments, the field is filled with zeros.
Theta	112	4	Binary Int.	Value of theta indicator from the current session. The indicator is calculated for options only. For other instruments, the field is filled with zeros.
Vega	116	4	Binary Int.	Value of vega indicator from the current session. The indicator is calculated for options only. For other instruments, the field is filled with zeros.
EndTrading	120	4	Binary Int.	End of trading date (YYYYMMDD). This field is filled in depending on the type of the instrument : • Bonds, mortgage-backed bonds: Date of bonds buy out • Futures: Contract expiration date

Field Name	Offset	Size (Bytes)	Format	Description
				• Options: Last trading day for options • Warrants: Last days if trading the warrants • Other instruments: Field completed with zeros
NumberOfInstruments	124	4	Binary Int.	Number of instruments admitted to trading For European type of warrants, the number of warrants in trade is equal to the number of issued warrants, while for the American type of warrants, the number of warrants in trade is equal to the number of issued warrants reduced by the number of executed warrants. For derivatives the field is completed with zeros.
TradingValueCurrency	128	8	Binary Int.	The field comprises the trading value expressed in trading currency. Trading value in currency is provided with fixed number of two decimal places. For instruments on which there were no trades concluded during the given session. The field is filled in with zeros.
InterestRate	136	4	Binary Int.	Interest rate is determined on the basis of WIBOR/WIBID rates for each expiration date of options and futures.

Field Name	Offset	Size (Bytes)	Format	Description
				Interest rate is used to determine Greek indicators (delta, gamma, rho, theta, vega). For other instruments, the field is filled with zeros.
Volatility	140	4	Binary Int.	Volatility to be calculated is based on the implied volatility (provided in field ImpliedVolatility) according to algorithm worked out by the WSE, which is available on the website: www.opcje.gpw.pl. This is the volatility which is taken for determination the Greek factors (delta, gamma, rho, theta, vega).
StrikePrice	144	4	Binary Int.	Strike price for option. Options' strike price is currently provided in the option series name. For the index option it is 8, 9,10 and 11 sign in the option series name. Example: WIG20 Index Options - OW20W143100 Strike price = 3100 For the stock options it is 8, 9, 10, 11, 12, 13 sign in the option series name. Example: KGH Stock Options - OKGHF15013025 Strike price = 130.25
ShortSellingTradesNumber	148	4	Binary Int.	Field should be ignored
VolumeShortSellingTrades	152	4	Binary Int.	Field should be ignored

Field Name	Offset	Size (Bytes)	Format	Description
ValueShortSellingTrades	156	4	Binary Int.	Field should be ignored
LastTradeDate	160	4	Binary Int.	The date of execution of the last trade for the given instrument (YYYYMMDD).
LastTradePrice	164	4	Binary Int.	The price of the last trade for given instrument. In case no trade were concluded in the trading session, the field contains the price of the last trade from the session, on which the given trade was concluded.
BlockTradesNumber	168	4	Binary Int.	The total number of block trades concluded on a particular instrument during the current trading session.
VolumeBlockTrades	172	4	Binary Int.	The total turnover volume of block trades concluded on a particular instrument during the current trading session.
ValueBlockTrades	176	4	Binary Int.	The total turnover value of block trades concluded on a particular instrument during the current trading session.
VWAPBlocktrade	180	4	Binary Int.	Volume weighted average price of block trades for a particular instrument during a particular trading session, calculated according the following algorithm: Σ(TCS trade price * TCS trade volume) / Σ (TCS trade volume) If no block trades were concluded during the trading session, the field is completed with zeros.
MaximumBlockPrice	184	4	Binary Int.	Maximum price of block trades in the instrument during the current trading session. If no block trades were concluded during the trading session, the field is completed with zeros.
MinimumBlockPrice	188	4	Binary Int.	Minimum price of block trades in the instrument during the current trading session. If no block trades were concluded during the trading session, the field is completed with zeros.
SystemID	192	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	196	2	Binary Int.	Number of micro seconds to be combined with SourceTime
MultiplierScaleCode	198	1	Binary Int.	Applicable to the multiplier.
AccumulatedInterestScaleCode	199	1	Binary Int.	Applicable to the accumulated interest.
OpeningPriceScaleCode	200	1	Binary Int.	Applicable to the opening price.
MaximumPriceScaleCode	201	1	Binary Int.	Applicable to the maximum price.
MinimumPriceScaleCode	202	1	Binary Int.	Applicable to the minimum price.
ClosingPriceScaleCode	203	1	Binary Int.	Applicable to the closing price.
ImpliedVolatilityScaleCode	204	1	Binary Int.	Applicable to the implied volatility.

Field Name	Offset	Size (Bytes)	Format	Description
AverageWeightedPriceScaleCode	205	1	Binary Int.	Applicable to the average weighted price.
ReferencePriceScaleCode	206	1	Binary Int.	Applicable to the reference price.
PctChangePriceScaleCode	207	1	Binary Int.	Applicable to the percentage change price.
DividendRateScaleCode	208	1	Binary Int.	Applicable to the dividend rate.
SettlementPriceScaleCode	209	1	Binary Int.	Applicable to the settlement price.
SettlementValueScaleCode	210	1	Binary Int.	Applicable to the settlement value.
Filler	211	1	*Binary Int.*	For future use.
TradingValueScaleCode	212	1	Binary Int.	Applicable to the trading value.
CurrencyPriceScaleCode	213	1	Binary Int.	Applicable to the currency price.
DeltaScaleCode	214	1	Binary Int.	Applicable to the delta.
GammaScaleCode	215	1	Binary Int.	Applicable to the gamma.
RhoScaleCode	216	1	Binary Int.	Applicable to the rho.
ThetaScaleCode	217	1	Binary Int.	Applicable to the theta.
VegaScaleCode	218	1	Binary Int.	Applicable to the vega.
TradingValueCurrencyScaleCode	219	1	Binary Int.	Applicable to the trading value.
InterestRateScaleCode	220	1	Binary Int.	Applicable to the interest rate.

Field Name	Offset	Size (Bytes)	Format	Description
VolatilityScaleCode	221	1	Binary Int.	Applicable to the volatility.
StrikePriceScaleCode	222	1	Binary Int.	Applicable to the strike price.
ValueShortSellingTradesScaleCode	223	1	Binary Int.	Field should be ignored
LastTradePriceScaleCode	224	1	Binary Int.	Applicable to the last trade price.
ValueBlockTradesScaleCode	225	1	Binary Int.	Applicable to the value of the block trades.
VWAPBlocktradeScaleCode	226	1	Binary Int.	Applicable to the VWAP block trade.
MaximumBlockPriceScaleCode	227	1	Binary Int.	Applicable to the maximum block price.
MinimumBlockPriceScaleCode	228	1	Binary Int.	Applicable to the minimum block price.
StatusIndicator	229	1	ASCII Str.	Defines the status of the instrument at the end of the stock exchange session. <u>Valid values:</u> 'Z' - Suspended Space - Active
Sector	230	3	ASCII Str.	Defines the sector of the economy that the company belongs to. Possible values for shares, the field assumes were described in the Appendix – 'Market Sectors'. For the remaining instruments the field is completed with spaces.

Field Name	Offset	Size (Bytes)	Format	Description
Market	233	1	ASCII Str.	Defines the market the instrument belongs to. Valid values: 'P' - Primary market 'R' - Parallel market 'Z' - New Connect Market (price driven market) 'C' - New Connect Market(order driven market) 'O' - Catalyst Regulated Market 'N' - Catalyst ASO
Filler	234	1	*Binary*	For future use
ShortSale	235	1	ASCII Str.	Field should be ignored
MarkerPctChangePrice	236	1	ASCII Str.	The field contains the market of the percentage change of the instrument closing price from the current session in relation to the reference price. Valid values: '+' - Increase '-' - Decrease Space - no change
SymbolDelta	237	1	ASCII Str.	The field contains a symbol of Delta indicator from the current session. Valid values: '+' - positive value '-' - negative value Space - zero value
SymbolGamma	238	1	ASCII Str.	The field contains a symbol of Gamma indicator from the current session. Valid values: '+' - positive value '-' - negative value Space - zero value
SymbolRho	239	1	ASCII Str.	The field contains a symbol of Rho indicator from the current session. Valid values: '+' - positive value '-' - negative value Space - zero value
SymbolTheta	240	1	ASCII Str.	The field contains a symbol of Theta indicator from the current session. Valid values: '+' - positive value '-' - negative value Space - zero value
SymbolVega	241	1	ASCII Str.	The field contains a symbol of Vega indicator from the current session. Valid values: '+' - positive value '-' - negative value Space - zero value
Filler	*242*	*1*	*Binary*	*For future use*

Field Name	Offset	Size (Bytes)	Format	Description
Filler	*243*	*1*	*Binary*	*For future use*
Filler	*244*	*1*	*Binary*	*For future use*
Filler	*245*	*1*	*Binary*	*For future use*
Filler	*246*	*1*	*Binary*	*For future use*
Filler	*247*	*1*	*Binary*	*For future use*
Segment	248	2	ASCII Str.	Informs in which segment the instrument is traded. Possible values for shares and allotment certificates are listed in Segments appendix.
LowerLiquidity	250	1	ASCII Str.	The field informs if the company was qualified to Lower Liquidity Space. For shares and rights to shares the field assumes one of the below-specified values:Valid values: 'T' - Company is contain in Lower Liquidity Space 'space' - remaining shares and other instruments
Dividend	251	2	ASCII Str.	This field informs that the instruments are traded the first time after determining the right to dividend. Valid values: 'bd' - without dividend
SubscriptionRight	253	2	ASCII Str.	This field informs that the securities are traded the first time after determining the subscription right. Valid values:
				'bp' - without subscription right
Split	255	2	ASCII Str.	This field informs that the securities are traded the first time after the change of the nominal value of shares. Valid values: 'pw' - After the change 'zj' - Change of the trading unit [starting from the moment of change until the last session before the date of the shares merger]
InterimDividendRight	257	2	ASCII Str.	This field informs that the shares are traded the first time after determining the interim dividend rights. Valid values: 'bz' - Without the interim dividend right
QuotationSystem	259	2	ASCII Str.	The field informing about the system of listing the instrument. Valid values: '81' - single price quotation system with a single fixing '82' - continuous trading '83' - single price quotation system with two fixings
LiquiditySupportPge	261	2	ASCII Str.	The field informs if the company entered to Liquidity Support Programme Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'T' - Company participates in Liquidity Support Programme 'spaces' - remaining shares and other instruments. As the length of this field is equal 2 the first position is filled in with value describing the segment and the second position is filled in with space.
Filler	*263*	*1*	*ASCII Str.*	*For future use.*

1.2.13 Generic WSE Message - 592 Message

1.2.13.1 Message Overview

This message is a free format message with a container of 250 bytes. WSE is responsible for defining the structure of these 250 bytes.

1.2.13.2 Message Sending Rules

The Generic WSE message with InternalMsgType =1 indicates the end of messages distribution for the current trading day. Message is sent at the end of the day.

The Generic WSE message with InternalMsgType =2 informs about the structure of the basket of treasury bonds for given bond future for the next trading day. Message is sent during the trading day.

The Generic WSE message with InternalMsgType =3 informs about the selected values for PRIIPs (recommended holding period) key information for the given trading day. It also includes the link to issuer's webside to the particular KID information for a given instrument. Message is sent at the beginning and end of the trading day.
The EU regulation no. 1286/2014 dated 26 November 2014 (PRIIPs regulation) came into effect on 31 December 2016 and it obligated the issuers to create and up to date basic information sheet (KID, Key Information Document) for PRIIPs (Packaged Retail and Insurance Based Investment Products). PRIIPs are packaged investment products with investment risks, whose value is derived from the reference titles and are part of several financial products.
The KIDs for PRIIPs should be handed over to potential investors before the conclusion of the contract. They should make it possible for the small investors in the EU to understand the basic features and risks of PRIIPs better. Moreover, the specifications for PRIIPs basic information sheets lead to such products being more comparable across Europe.

The Generic WSE message with InternalMsgType =4 informs about the beginning or the end of Stressed Market Conditions State for a given instrument. Message is sent during the trading day.

The Generic WSE message with InternalMsgType =5 informs about the beginning or the end of Exceptional Market Conditions State for a given instrument. Message is sent during the trading day.

1.2.13.3 Message Structure

The table below describes the body fields of a Generic WSE message (MsgType = '592').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	592 – Generic WSE message

Field Name	Offset	Size (Bytes)	Format	Description
SourceSeqNum	4	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SystemID	12	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	16	2	Binary Int.	Number of micro seconds. To be combined with SourceTime.
SourceData	18	1	ASCII Str	Source of Data: '1' - UTP Cash '2' - UTP Derivatives '3' - Indexator '4' -BondSpot '5' – other 'space' - not applicable
InternalMsgType	19	2	ASCII Str.	This field specifies the internal message type. Valid values: 01 – End of the session 02 – Treasury Bond Futures Data 03 – KID
				04 – Stressed Market Conditions 05 – Exceptional Market Conditions
EventDate	21	8	ASCII Str.	Date of the event YYYYMMDD
InstrumentID	29	12	ASCII Str.	WSE: TradingCode BS: InstrumentCode 'Space' – not applicable
InstrumentGroupCode	41	3	ASCII Str.	Designates the class to which the security belongs.
MsgTypeIndicator	44	1	ASCII Str.	Field defines the character or nature of the message being sent. Valid values: The field assumes one of the below specified values: 'F' – functional 'T' – technical 'Space' - not applicable
LanguageVersion	45	1	ASCII Str.	Field defines the language of the contents of the message. Valid values: 'P' – Polish version of the message 'E' – English version of the message 'Space' – Not applicable
TypeOffRecipient	46	1	ASCII Str.	This field defines the class of the recipient of a message being sent This field assumes one of the below specified values: 'D' – distributors of information 'M' – exchange members

Field Name	Offset	Size (Bytes)	Format	Description
				'A' – all recipients
MsgSerialNum	47	1	ASCII Str.	The message serial number within the framework of the whole given message based on InternalMsgType
BodyMessage	48	220	ASCII Str.	Filler – when **InternalMsgType = 01** Below defined data – for next **InternalMsgTypes (02,03,….)**

Body Message for **InternalMsgType = 02**

Body message	ISIN Code	48	12	ASCII Str.	Treasury Bond Future ISIN Code
	Date of first quotation	60	8	ASCII Str.	Date of first quotation of given futures (indicated in field ISIN Code)
	Short Name	68	16	ASCII Str.	Treasury Bond Future short name
	Date	84	8	ASCII Str.	Start date
	Underlying ISIN Code	92	12	ASCII Str.	Underlying ISIN Code for specified Futures Class
	(1) Treasury Bond ISIN Code	104	12	ASCII Str.	ISIN Code of (1) Treasury Bond series included in the bond basket.
	(1) Conversion Coefficient factor	116	9 [3,6]	ASCII Str.	Conversion coefficient value for (1) Treasury Bonds series. Numeric value with a fixed decimal places =6
	(2) Treasury Bond ISIN Code	125	12	ASCII Str.	ISIN Code of (2) Treasury Bond series included in the bond basket.
	(2) Conversion Coefficient factor	137	9 [3,6]	ASCII Str.	Conversion coefficient value for (2) Treasury Bonds series. Numeric value with a fixed decimal places =6
	(3) Treasury Bond ISIN Code	146	12	ASCII Str.	ISIN Code of (3) Treasury Bond series included in the bond basket.
	(3) Conversion Coefficient factor	158	9 [3,6]	ASCII Str.	Conversion coefficient value for (3) Treasury Bonds series. Numeric value with a fixed decimal places =6
	(4) Treasury Bond ISIN Code	167	12	ASCII Str.	ISIN Code of (4) Treasury Bond series included in the bond basket. Filler if not used.
	(4) Conversion Coefficient factor	179	9 [3,6]	ASCII Str.	Conversion coefficient value for (4) Treasury Bonds series. Numeric value with a fixed decimal places =6 Filler if not used.
	(5) Treasury Bond ISIN Code	188	12	ASCII Str.	ISIN Code of (5) Treasury Bond series included in the bond basket. Filler if not used.
	(5) Conversion Coefficient factor	200	9 [3,6]	ASCII Str.	Conversion coefficient value for (5) Treasury Bonds series. Numeric value with a fixed decimal places =6Filler if not used.
	Filler	209	59	ASCII Str.	

Body Message **for InternalMsgType field = 03**

Body message	ISIN Code	48	12	ASCII Str.	ISIN Code
	Type of Instrument	60	4	ASCII Str.	Type of instrument / Categorization code
	KID version	64	10	ASCII Str.	Specifies the version of KID In case of first publication == date of creation of KID Possible value format: YYYYMMDDNN i.e 2017040100
	KID Issue date	74	8	ASCII Str.	Date of creation and first publication of KID for given instrument / Issue date
	Value At Risk	82	1	ASCII Str.	/ description/ Value included in this field depends on the information delivered
	Holding Period	83	3	ASCII Str.	Specifies the PRIIPs' recommended holding period. Value included in this field depends on the information delivered by an Issuer. WSE has no liability for the value. D01-D099 M01-M12 Y01-Y05
	PRIIP KID/url	86	182	ASCII Str.	Specifies the URL from which the KID can be retrieved. Value included in this field depends on the information delivered by an Issuer. WSE has no liability for the value.

***

Zastrzeżenie prawne

Należy zwrócić uwagę, iż dane dotyczące KID pochodzą ze źródeł zewnętrznych i nie były niezależnie weryfikowane. W odniesieniu do wyczerpującego charakteru lub rzetelności informacji zawartych w komunikatach, nie mogą być udzielone żadne zapewnienia ani oświadczenia.

GPW nie ponosi żadnej odpowiedzialności za jakiekolwiek decyzje podjęte w oparciu o informacje zawarte w linku do strony emitenta.

***

Disclaimer

Data regarding KIDs are derived from external sources and have not been independently verified. No warranty or representation can be given that information in the messages are exhaustive or true.

The WSE has no liability for any decisions made on the basis of any information or opinion in links to issuer's web sides.

Body Message for **InternalMsgType field = 04, 05**

Body message	ISIN Code	48	12	ASCII Str.	ISIN Code
	Market Conditions State	60	1	ASCII Str.	Indicates the state of market conditions Valid values: Y – Beginning of Market Conditions State Z – End of Market Conditions State
	Market Conditions Number	61	4	ASCII Str.	Unique numeric and increasing identifier of Market Conditions during a day
	Filler	65	203	ASCII Str.	Filler

1.2.14 Open Position - 595 Message

1.2.14.1 Message Overview

The Open Position message indicates the current number of open positions for a particular instrument.

1.2.14.2 Message Sending Rules

This message is sent:

- After each transaction concluded in the derivatives market
- After the end of the trading session
- Before the start of the trading session

1.2.14.3 Message Structure

The table below describes the body fields of an Open Position message (MsgType = '595').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	595 – Instrument open position
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day.

Field Name	Offset	Size (Bytes)	Format	Description
				Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
TradingDate	16	8	ASCII Str.	Date of trading day (YYYYMMDD).
OpenPosition	24	4	Binary Int.	Number of open positions
TransactionNumber	28	4	Binary Int.	Number of transactions when OpenPositionIndicator = 0. it is a unique number of transaction for a particular security on a given Stock Exchange trading day; otherwise, the field is filled in with zeros.
SystemID	32	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	36	2	Binary Int.	Number of micro seconds to be combined with SourceTime
OpenPositionIndicator	38	1	ASCII Str.	The field defines whether: -the NOP value being sent is a value calculated during the quotation based on the incoming TRANSACTION messages – and thus an estimated value -the NOP value being sent is the last on a given day -the value being sent is a verified value Valid values: '0' - Open position during the quotation phase '3' - Open position after the end of the trading session '5' - Open position before the start of the trading session (verified data from the previous day)
Filler	*39*	*1*	*Binary*	*For future use*

1.3 PRICES AND TRADES

1.3.1 Overview

The Prices and Trades service uses the push-based publishing model. This means that data will be published based on its availability. Once an update on a price or on a trade is available, it will be published to Trades Clients.

The Trade message reflects the last sale in each traded security.

Below is a list of the functional message types in the Prices and Trades Feed:

- 240 - Trade Full Information – 276 bytes in total, **274** excluding MsgSize field
- 241 - Price Update – 44 bytes in total, **42** excluding MsgSize field
- 243 – Trade publication - 288 bytes in total, **286** excluding MsgSize field
- 245 - Auction Summary – 52 bytes in total, **50** excluding MsgSize field
- 247 – VWAP-Closing Price – 56 bytes in total, **54** excluding MsgSize field

Below is a list of the functional message types for BondSpot in the Prices and Trades Feed:

- 240 - Trade Full Information – 276 bytes in total, **274** excluding MsgSize field
- 241 - Price Update – 44 bytes in total, **42** excluding MsgSize field

1.3.2 Trade Full Information - 240 Message

New Mifid II fields have been added to the message.

1.3.2.1 Message Overview

Trade full information message is sent each time a trade or a cancellation of a trade has occurred. This specific message provides the detailed information related to the trade.

1.3.2.2 Message Sending Rules

This message is sent each time a trade is taking place on the trading engine.

This message is also sent for a trade cancellation. When the field MMTModificationIndicator is set to 'CANC' it will identify the case of a trade being cancelled while the other fields remain the same as the original trade.

1.3.2.3 Message Structure

The table below describes the body fields of a Trade Full Information message (MsgType = '240').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field
MsgType	2	2	Binary Int.	240 – Trade Full Information
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the Trade generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170

Field Name	Offset	Size (Bytes)	Format	Description
TradeIDNumber	12	4	Binary Int.	Unique numeric and increasing Identifier of the trade, set by the trading engine.
VenueOfExecution (new field instead of QuoteLinkID that was not used)	16	4	ASCII Ch.	Identification of the venue where the transaction was executed. ISO 10383 segment MIC for transactions executed on a trading venue. Where the segment MIC does not exist, use the operating MIC.
SourceSeqNum	20	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
PriceNotation (new field instead of Price)	24	4	ASCII Ch.	Indication as to whether the price is expressed in monetary value, in percentage or in yield. Possible values: 'MONE' : monetary value 'PERC' : percentage 'YIEL' : yield 'BAPO' : basis points
TransactionToBeCleared (new field instead of Volume)	28	4	ASCII Ch.	Code to identify whether the transaction will be cleared. Always equal to 'TRUE'
CumulativeQuantity	32	4	Binary Int.	Cumulative number of securities traded on UTP since the start of the current trading session Note: Value accumulated within the individual segments
HighestPrice	36	4	Binary Int.	Highest Price traded during the day(To be calculated with the PriceScaleCode)
LowestPrice	40	4	Binary Int.	Lowest Price traded during the day(To be calculated with the PriceScaleCode)

Field Name	Offset	Size (Bytes)	Format	Description
VariationLastPrice	44	4	Binary Int. (signed)	Percentage variation of Last Traded Price to last reference price (To be calculated with the VariationScaleCode)
SystemID	48	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	52	2	Binary Int.	Number of micro seconds. To be combined with SourceTime field.
PriceCurrency (new field instead of filler)	54	3	ASCII Ch.	Currency in which the price is expressed as defined by ISO 4217 (applicable if the price is expressed as monetary value).
LastTradeIndicator Field name changed only (previous TradCond2)	57	1	ASCII Ch.	Last trade at same price indicator: Valid values: '0' - Not last of a series of trade at the same price. '1' - Last of a series of trade at the same price BondSpot: Not applicable, always set to '1'
TypeOfTrade Field name changed only (previous TradCond3)	58	1	ASCII Ch.	Trade related conditions. Cross Trade Indicator: Valid values: '0' - Trade does not stem from a Cross Order '1' - Trade stems from a Cross Order '4' - Valuation Trade '5' - Internalized Trade '6' - Internalized Cross Trade
TradeOrigin	59	1	ASCII Ch.	This field indicates the origin of the trade. Valid values: 'B' - Orders from the book 'O' - Block trade 'I' - IPO trade 'T' - Tender offer trade 'S' - BISO trade 'R' - Trade reporting (for future use)
TickDirection	60	1	ASCII Ch.	Symbol of the variation of the price versus the previous (traded or reference) price. Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'+' - positive '-' - negative '0' - no variation or data field not significant (used for trade cancellation)
OpeningTradeIndicator	61	1	ASCII Ch.	If the trade took place during the opening auction or during the core session. Valid values: 'O' - Opening 'S' - Core Continuous BondSpot: always filled with 'space'
VariationScaleCode	62	1	Binary Int.	To be combined with VariationLastPrice
PriceScaleCode	63	1	Binary Int.	Applicable to all prices in the message except for the field "Price"
TradingDateTime (new field instead of SS Trade)	64	27	ASCII Ch.	Date and time when the transaction was executed. Date and time in the following format: YYYY-MM-DDThh:mm:ss.dddddd Z. – 'YYYY' is the year – 'MM' is the month – 'DD' is the day – 'T' – means that the letter 'T' shall be used – 'hh' is the hour – 'mm' is the minute – 'ss.dddddd' is the second and its fraction of a second – Z is UTC time Dates and times shall be reported in UTC.
PublicationDateTime	91	27	ASCII Ch.	Date and time when the transaction was published by a trading venue. Date and time in the following format: YYYY-MM-DDThh:mm:ss.dddddd Z. – 'YYYY' is the year

Field Name	Offset	Size (Bytes)	Format	Description
				– 'MM' is the month – 'DD' is the day – 'T' means that the letter 'T' shall be used – 'hh' is the hour – 'mm' is the minute – 'ss.dddddd' is the second and its fraction of a second – Z is UTC time Dates and times shall be reported in UTC.
InstrumentIdCodeType	118	4	ASCII Ch.	Code type used to identify the financial instrument Always equal to 'ISIN'
InstrumentIdCode	122	12	ASCII Ch.	ISIN code used to identify the financial instrument, as defined in ISO 6166.
Price (new field format changed)	134	20	ASCII Ch.	Traded price of the transaction excluding, where applicable, commission and accrued interest. In the case of option contracts, it shall be the premium of the derivative contract per underlying or index point. Where price is reported in monetary terms, it shall be provided in the major currency unit. The information reported in this field shall be consistent with the values provided in field Quantity. Possible values:

Field Name	Offset	Size (Bytes)	Format	Description
				Price format depends on the 'Price Notation' field. – if price is expressed as a monetary value, maximum of 18 digits with a maximum of 13 decimals – if price is expressed as percentage or yield, maximum of 11 digits with a maximum of 10 decimals – if price is expressed as basis points, maximum of 18 digits with a maximum of 17 decimals Where applicable the following format should be considered: note 1: decimal separator is '.' (full stop) note 3: where applicable, values shall be rounded and not truncated note 4: the value will be padded on the left side with '0' (zero)
Quantity (new field format changed)	154	20	ASCII Ch.	Number of units of the financial instrument, or the number of derivative contracts in the transaction. The nominal or monetary value of the financial instrument. Possible values: – if quantity is expressed as number of units, maximum of 18 digits with a maximum of 17 decimals Where applicable the following format should be considered: note 4: the value will be padded on the left side with '0' (zero)
NotionalAmount (new field)	174	20	ASCII Ch.	Nominal amount or notional amount. The information reported in this field shall be consistent with the value provided in field Price.

Field Name	Offset	Size (Bytes)	Format	Description
				Possible values: – maximum of 18 digits with a maximum of 5 decimals Where applicable the following format should be considered: note 1: decimal separator is '.' (full stop) note 3: where applicable, values shall be rounded and not truncated note 4: the value will be padded on the left side with '0' (zero)
NotionalCurrency (new field)	194	3	ASCII Ch.	Currency in which the notional is denominated as defined by ISO 4217.
Filler	197	1	Binary	For future use
TransactionIDCode (new field)	198	52	ASCII Ch.	Alphanumerical code assigned by trading venue used in any subsequent reference to the specific trade. The transaction identification code shall be unique, consistent and persistent per ISO10383 segment MIC and per trading day. Where the trading venue does not use segment MICs, the transaction identification code shall be unique, consistent and persistent per operating MIC per trading day. Transaction Identification Code is composed of the Symbol, Venue of Execution and the TradeIDNumber / TCSTradeId to make it unique.
MMTMarketMechanism (new field)	250	1	ASCII Ch.	Defines the market mechanism following MMT level 1. Possible values: '1' : Central Limit Order Book '2' : Quote Driven Market (not used) '3' : Dark Order Book (not used) '4' : Off Book (including Voice or Messaging Trading) '5' : Periodic Auction (not used) '6' : Request for Quotes (not used)

Field Name	Offset	Size (Bytes)	Format	Description
MMTTradingMode (new field)	251	1	ASCII Ch.	Defines the trading Mode mechanism following MMT level 2 Possible values: - '1': Undefined Auction (not used) - 'O': Scheduled Opening Auction - 'K': Scheduled Closing Auction - 'I': Scheduled Intraday Auction (not used) - 'U': Unscheduled Auction - '2': Continuous Trading - '3': At Market Close Trading - '4': Out of Main Session Trading (not used) - '5': Trade Reporting (On Exchange) - '6': Trade Reporting (Off Exchange) - '7': Trade Reporting (Systematic Internaliser) (not used)
MMTAlgorithmicIndicator (new field)	252	4	ASCII Ch.	Defines the algorithmic indicator following MMT level 3.9. Possible values: - 'ALGO': Algorithmic Trade - '-': No Algorithmic Trade Definitions: ALGO – transactions executed as a result of an investment firm engaging in algorithmic trading as defined in Article 4(1)(39) of Directive 2014/65/EU.
MMTModificationIndicator	256	4	ASCII Ch.	Defines the modification indicator following MMT level 3.4. Possible values: - 'CANC': Trade Cancellation - 'AMND': Trade Amendment (not used) - '-': New Trade Definitions: CANC – when a previously published transaction is cancelled AMND – when a previously published transaction is amended
MMTNegotiationIndicator (new field)	260	4	ASCII Ch.	Defines the negotiation indicator or pre-trade transparency waiver following MMT level 3.2.
				Possible values: - 'N': Negotiated Trade (not used) - 'NLIQ': Negotiated Trade in Liquid Financial Instruments - 'OILQ': Negotiated Trade in Illiquid Financial Instruments - 'PRIC': Negotiated Trade Subject to Conditions Other Than The Current Market Price (not used) - '-': No Negotiated Trade Definitions: N – [to be defined] NLIQ – transactions executed in accordance with Article 4(1)(b)(i) of Regulation (EU) No 600/2014 OILQ – transactions executed in accordance with Article 4(1)(b)(ii) of Regulation (EU) No 600/2014 PRIC – transactions executed in accordance with Article 4(1)(b)(iii) of Regulation (EU) No 600/2014 and as set out in Article 6 of Regulation [RTS 1]
MMTSpecialDividendIndicator (new field)	264	4	ASCII Ch.	Defines the special dividend indicator following MMT level 3.6. Possible values: - 'SDIV': Special Dividend Trade - '-': No Special Dividend Trade Definitions: SDIV – transactions that are either: – executed during the ex-dividend period where the dividend or other form of distribution accrues to the buyer instead of the seller, or – executed during the cum-dividend period where the dividend or other form of distribution accrues to the seller instead of the buyer
MMTContributionToPrice (new field)	268	4	ASCII Ch.	Defines the contribution to price or the price discovery process following MMT level 3.8. Possible values: - 'P': Plain-Vanilla Trade - 'NPFT': Non-Price Forming Trade (formerly known as the Technical Trade)

Field Name	Offset	Size (Bytes)	Format	Description
				- 'TNCP': Trade not Contributing to the Price Discovery Process (not used) Definitions: NPFT – transactions where the exchange of financial instruments is determined by factors other than the current market valuation of the financial instrument as listed under Article 13 of Regulation [RTS 1] and Article 12 of Regulation [RTS 2] TNCP – transactions not contributing to the price discovery process for the purposes of Article 23 of Regulation (EU) No 600/2014 and as set out in Article 2 of Regulation [RTS 1]
MMTBenchmarkIndicator (new field)	272	4	ASCII Ch.	Defines the benchmark indicator or the reference price indicator following MMT level 3.5. Possible values: - 'BENC': Benchmark Trade (not used) - 'RFPT': Reference Price Trade (not used) - '-': No Benchmark or Reference Price Trade Definitions: BENC – transactions executed in reference to a price that is calculated over multiple time instances according to a given benchmark, such as volume-weighted average price or time-weighted average price RFPT – transactions which are executed under systems operating in accordance with Article 4(1)(a) of Regulation (EU) No 600/2014

1.3.3 Price Update - 241 Message

1.3.3.1 Message Overview

The Price update message contains a bid price or an ask price in case of alternative indicative price (AIP), or the modification of the last reference price on a given instrument. This message informs of:

- Modifications operated on instrument prices.
- When the message follows the cancellation of a trade, it indicates that an instrument's most recent price has been modified.
- The updated instrument's last adjusted closing price or the instrument's settlement or theoretical price when modified.

1.3.3.2 Message Sending Rules

This message is sent:

- When the market operation changes the last adjusted price of the previous day (type of price 34).
- When the market operation cancels a trade which modifies the last traded price (type of price 33).
- When a member enters an order on an empty order book on a Class where the AIP Type is configured “Order Entry” (type of price: 02, 03).
- When the derivatives instrument daily settlement price (type of price 35) or final settlement price (type of price 36) or theoretical price (type of price 37) is known or updated.
- When an member firm’s AIP Price Input message occurs on an instrument that is configured with AIP Type equals to “Member Firm Price” (type of price: 51).
- On the first trade of the day (type of price 30) or after a trade cancellation which updates the value of the first traded price (type of price 30). If the trade cancellation makes there is no first trade anymore on the instrument, the output price message is generated with a non significative value in price.

- For instruments (IN and LE market segments), the level 34 of TypeOfPrice field indicates that the disseminated price is either the reference price (if there were no LP quotes submitted so far, e.g. on the first day of trading) or the last known LP midpoint otherwise i.e. (LP Bid + LP Offer)/2.

1.3.3.3 Message Structure

The table below describes the body fields of a Price Update message (MsgType = '241').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	241 - Price Update
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
Price	16	4	Binary Int.	Last Price of the trading day (To be calculated with the ScaleCode)
HighestPrice	20	4	Binary Int.	Highest Price traded during the day (To be calculated with the PriceScaleCode)
LowestPrice	24	4	Binary Int.	Lowest Price traded during the day (To be calculated with the PriceScaleCode)
VariationLastPrice	28	4	Binary Int. (signed)	Percentage variation of the price / last reference price (To be calculated with the VariationScaleCode)
SystemID	32	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	36	2	Binary Int.	Number of micro seconds. To be combined with SourceTime field.
TypeOfPrice	38	2	Binary Int.	Market- Code identifying the type of price that is updated. Valid values: 02 - Bid price 03 - Ask price 30 - First traded price of the day 33 - New last price 34 - New previous day's closing price (Last Adjusted Closing Price or LP midpoint) 35 - Daily settlement price 36 - Final settlement price 37 - Theoretical price 51 - Alternative Indicative Price AIP For BondSpot: 02 - Bid price – indicative sell price of instrument calculated in accordance with "Regulation for treasury Securities Fixing" determined by NBP (National Bank of Poland)

Field Name	Offset	Size (Bytes)	Format	Description
				03 - Ask price - Indicative buy price of instrument calculated in accordance with "Regulation for treasury Securities Fixing" determined by NBP (National Bank of Poland) 30 - Change of first price 31 - New high price 32 - New low price 33 - New last price 34 - Reference price For BondSpot Regulated Market and Alternative Trading System stream price is calculated in the following way: - Price of the last trade; - In case of lack of trade – issue price. For TBSP stream the price is calculated in the following way: - Last TBSP.fixPrice - In case of lack of TBSP.fixprice minimum auction price from the last auction. 50 - Fixing price – for given instrument is calculated as the arithmetic mean of 'indicative buy price' and 'indicative sell price' of a given treasury security In case of lack of fixing price in messages 241 for "TypeOfPrice"= "02", "03", "50" fields "Price", „HighestPrice", „LowestPrice" are populated with value „2808348671" and the field „VariationLastPrice" with value „- 1486618625". Messages with values mentioned above should be ignored.

Field Name	Offset	Size (Bytes)	Format	Description
				77 - Reference Price TBSP.Price 78 - Reference Price TBSP.fixPrice 88 - MidPrice 99 – Treasury BondSpot Spot Price – reference price for REPO trades for fixing instruments - fixing price calculated on the second fixing session. In case of lack of fixing price – clear reference spot price calculated in accordance with "Regulation for Treasury Securities Fixing" of BondSpot. In case of lack of price fields "Price", „HighestPrice", „LowestPrice" are populated with value „2808348671" and the field „VariationLastPrice" with value „- 1486618625". Messages with values mentioned above should be ignored. -
Filler	40	*2*	*Binary*	*For future use.*
PriceScaleCode	42	1	Binary Int.	Applicable to all prices in the message
VariationScaleCode	43	1	Binary Int.	Available for the price variation in the message

1.3.4 Trade Publication - 243 Message

Note

This message is not used.

New Mifid II fields have been added to the message.

1.3.4.1 Message Overview

A trade publication message is sent by the reporting system whenever a trade is reported. A trade publication message is sent through the Market Data dissemination flows if and at the moment that it is defined for publication.

This message is either sent for a trade creation or cancellation.

1.3.4.2 Message Structure

The table below describes the body fields of a Trade Publication message (MsgType = '243').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	243 – Trade Publication
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the Trade generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit
VenueOfExecution (new field instead of price)	16	4	ASCII Ch.	Identification of the venue where the transaction was executed. ISO 10383 segment MIC for transactions executed on a trading venue. Where the segment MIC does not exist, the operating MIC.
InstrumentIdCodeType (new field instead of quantity)	20	4	ASCII Ch.	Code type used to identify the financial instrument Always equal to 'ISIN'
PriceMultiplier	24	4	Binary Int.	Number of units of the financial instrument that are contained a trading lot (to be calculated with the MultiplierScaleCode)
SystemID	28	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	32	2	Binary Int.	Number of micro seconds. To be combined with SourceTime field.
TradingDateTime (new field)	34	27	ASCII Ch.	Date and time when the transaction was executed. Date and time in the following format: YYYY-MM-DDThh:mm:ss.ddddddZ. – 'YYYY' is the year – 'MM' is the month – 'DD' is the day – 'T' – means that the letter 'T' shall be used – 'hh' is the hour – 'mm' is the minute

Field Name	Offset	Size (Bytes)	Format	Description
				– 'ss.dddddd' is the second and its fraction of a second – Z is UTC time Dates and times shall be reported in UTC.
PriceCurrency (new field)	61	3	ASCII Ch.	Currency in which the price is expressed (applicable if the price is expressed as monetary value).
InstrumentIdCode (new field)	64	12	ASCII Ch.	ISIN code used to identify the financial instrument, as defined in ISO 6166. ISIN Code used to identify the financial instrument
PublicationDateTime (new field)	76	27	ASCII Ch.	Date and time when the transaction was published by a trading venue Date and time in the following format: YYYY-MM-DDThh:mm:ss.ddddddZ. - 'YYYY' is the year; - 'MM' is the month; - 'DD' is the day; - 'T' – means that the letter 'T' shall be used - 'hh' is the hour; - 'mm' is the minute; - 'ss.dddddd' is the second and its fraction of a second; - Z is UTC time. Dates and times shall be reported in UTC.
NotionalCurrency (new field)	103	3	ASCII Ch.	Currency in which the notional is denominated as defined by ISO 4217.
TCSTradeID	106	16	ASCII Ch.	Unique reference of the trade
PriceNotation	122	4	ASCII Ch.	Indication as to whether the price and the strike price is expressed in monetary value, in percentage or in yield

Field Name	Offset	Size (Bytes)	Format	Description
				Possible values: - MONE: monetary value. - PERC: percentage. - YIEL: yield. - BAPO: basis points
Fillller	126	1	*Binary*	*For future use*
TransactionToBeCleared (new field)	127	4	ASCII Ch.	Code to identify whether the transaction will be cleared.
MMTBenchmarkIndicator (new field)	131	4	ASCII Ch.	Defines the benchmark indicator or the reference price indicator following MMT level 3.5. Possible values: - 'BENC': Benchmark Trade (not used) - 'RFPT': Reference Price Trade (not used) - '-': No Benchmark or Reference Price Trade Definitions: BENC – transactions executed in reference to a price that is calculated over multiple time instances according to a given benchmark, such as volume-weighted average price or time-weighted average price RFPT – transactions which are executed under systems operating in accordance with Article 4(1)(a) of Regulation (EU) No 600/2014
MMTContributionToPrice (new field)	135	4	ASCII Ch.	Defines the contribution to price or the price discovery process following MMT level 3.8. Possible values: - 'P': Plain-Vanilla Trade - 'NPFT': Non-Price Forming Trade (formerly known as the Technical Trade) - 'TNCP': Trade not Contributing to the Price Discovery Process (not used) Definitions:

Field Name	Offset	Size (Bytes)	Format	Description
				NPFT – transactions where the exchange of financial instruments is determined by factors other than the current market valuation of the financial instrument as listed under Article 13 of Regulation [RTS 1] and Article 12 of Regulation [RTS 2] TNCP – transactions not contributing to the price discovery process for the purposes of Article 23 of Regulation (EU) No 600/2014 and as set out in Article 2 of Regulation [RTS 1]
MultiplierScaleCode	139	1	Binary Int.	To be combined with PriceMultiplier
Price (new field)	140	20	ASCII Ch.	Traded price of the transaction excluding, where applicable, commission and accrued interest. In the case of option contracts, it shall be the premium of the derivative contract per underlying or index point. Where price is reported in monetary terms, it shall be provided in the major currency unit. The information reported in this field shall be consistent with the values provided in field Quantity. Possible values: Price format depends on the 'Price Notation' field. – if price is expressed as a monetary value, maximum of 18 digits with a maximum of 13 decimals – if price is expressed as percentage or yield, maximum of 11 digits with a maximum of 10 decimals – if price is expressed as basis points, maximum of 18 digits with a maximum of 17 decimals Where applicable the following format should be considered: note 1: decimal separator is '.' (full stop)
				note 2: negative numbers are prefixed with '-' (minus) note 3: where applicable, values shall be rounded and not truncated note 4: the value will be padded on the left side with '0' (zero)
Quantity (new field)	160	20	ASCII Ch.	Replacing Volume Number of units of the financial instrument. The nominal or monetary value of the financial instrument. Possible values: - For quantity expressed as number of units: maximum of 18 digits with a maximum of 17 decimals. Where applicable the following format should be considered: note 1: decimal separator is '.' (full stop) note 2: negative numbers are prefixed with '-' (minus) note 3: where applicable, values shall be rounded and not truncated note 4: the value will be padded on the left side with '0' (zero)
NotionalAmount (new field)	180	20	ASCII Ch.	Nominal amount or notional amount. The information reported in this field shall be consistent with the value provided in field Price. Possible values: – maximum of 18 digits with a maximum of 5 decimals Where applicable the following format should be considered: note 1: decimal separator is '.' (full stop) note 2: negative numbers are prefixed with '-' (minus) note 3: where applicable, values shall be rounded and not truncated note 4: the value will be padded on the left side with '0' (zero)
TransactionIDCode (new field)	200	52	ASCII Ch.	Alphanumerical code assigned by trading venue used in any subsequent reference to the specific trade.

Field Name	Offset	Size (Bytes)	Format	Description
				The transaction identification code shall be unique, consistent and persistent per ISO10383 segment MIC and per trading day. Where the trading venue does not use segment MICs, the transaction identification code shall be unique, consistent and persistent per operating MIC per trading day. Transaction Identification Code is composed of the Symbol, Venue of Execution and the TradeIDNumber / TCSTradeId to make it unique.
MMTMarketMechanism (new field)	252	1	ASCII Ch.	Defines the market mechanism following MMT level 1. Possible values: '1' : Central Limit Order Book '2' : Quote Driven Market (not used) '3' : Dark Order Book (not used) '4' : Off Book (including Voice or Messaging Trading) '5' : Periodic Auction (not used) '6' : Request for Quotes (not used)
MMTTradingMode (new field)	253	1	ASCII Ch.	Defines the trading Mode mechanism following MMT level 2 Possible values: - '1': Undefined Auction (not used) - 'O': Scheduled Opening Auction - 'K': Scheduled Closing Auction - 'I': Scheduled Intraday Auction (not used) - 'U': Unscheduled Auction - '2': Continuous Trading - '3': At Market Close Trading - '4': Out of Main Session Trading (not used) - '5': Trade Reporting (On Exchange) - '6': Trade Reporting (Off Exchange) - '7': Trade Reporting (Systematic Internaliser) (not used) - '-': None apply
MMTSpecialDividendIndicator	254	4	ASCII Ch.	Defines the special dividend indicator following MMT level 3.6. Possible values: - 'SDIV': Special Dividend Trade - '-': No Special Dividend Trade Definitions: SDIV – transactions that are either: – executed during the ex-dividend period where the dividend or other form of distribution accrues to the buyer instead of the seller, or – executed during the cum-dividend period where the dividend or other form of distribution accrues to the seller instead of the buyer
MMTNegotiationIndicator (new field)	258	4	ASCII Ch.	Defines the negotiation indicator or pre-trade transparency waiver following MMT level 3.2. Possible values: - 'N': Negotiated Trade (not used) - 'NLIQ': Negotiated Trade in Liquid Financial Instruments - 'OILQ': Negotiated Trade in Illiquid Financial Instruments - 'PRIC': Negotiated Trade Subject to Conditions Other Than The Current Market Price (not used) - '-': No Negotiated Trade Definitions: N – [to be defined] NLIQ – transactions executed in accordance with Article 4(1)(b)(i) of Regulation (EU) No 600/2014 OILQ – transactions executed in accordance with Article 4(1)(b)(ii) of Regulation (EU) No 600/2014 PRIC – transactions executed in accordance with Article 4(1)(b)(iii) of Regulation (EU) No 600/2014 and as set out in Article 6 of Regulation [RTS 1]
MMTAgencyCrossIndicator (New field)	262	4	ASCII Ch.	Transactions where an investment firm has brought together clients' orders with the purchase and the sale conducted as one transaction and involving the same volume and price. Possible values: - 'ACTX' Agency Cross Trade

Field Name	Offset	Size (Bytes)	Format	Description
				- '-' No Agency Cross Trade
MMTTransactionCategory (new field)	266	4	ASCII Ch.	Defines the transaction category following MMT level 3.1. Possible values: - 'D': Dark Trade (not used) - 'RPRI': Trade that has Received Price Improvement (not used) - 'TPAC': Package Trade (excluding Exchange for Physicals) (not used) - 'XFPH': Exchange for Physicals Trade (not used) - '-': None apply (a standard trade for the Market Mechanism and Trading Mode) Definitions: RPRI – transactions executed on a systematic internaliser with a price improvement in accordance with Article 15(2) of Regulation (EU) No 600/2014 TPAC – package transactions which are not exchange for physicals as defined in Article 1 of Regulation [RTS 2] XFPH – exchange for physicals as defined in Article 1 of Regulation [RTS 2]
MMTModificationIndicator (new field)	270	4	ASCII Ch.	Defines the modification indicator following MMT level 3.4. Possible values: - 'CANC': Trade Cancellation - 'AMND': Trade Amendment (not used) - '-': New Trade Definitions: CANC – when a previously published transaction is cancelled AMND – when a previously published transaction is amended
MMTDuplicativeIndicator (new field)	274	4	ASCII Ch.	Defines the duplicative indicator following MMT level 5. Possible values: - 'DUPL': Duplicative Trade Report (reported to more than one APA) - '-': Unique Trade Report
				Definitions: DUPL – when a transaction is reported to more than one APA in accordance with Article 17(1) of Commission Delegated Regulation (EU) on [DRSP RTS]
MMTPublicationMode (new field)	278	4	ASCII Ch.	Defines the publication mode or post-trade deferral reason following MMT level 4.1. Possible values: - '-': Immediate Publication - '1': Non-Immediate Publication (not used) - 'LRGS': Non-Immediate Publication: Deferral for "Large in Scale" (not used) - 'ILQD': Non-Immediate Publication: Deferral for "Illiquid Instrument" (not used) - 'SIZE': Non-Immediate Publication: Deferral for "Size Specific" (not used) Definitions: 1 – [to be defined] LRGS – transactions executed under the post-trade large in scale deferral ILQD – transactions executed under the deferral for instruments for which there is not a liquid market SIZE – transactions executed under the post-trade size specific to the instrument deferral
MMTAlgorithmicIndicator (new field)	282	4	ASCII Ch.	Defines the algorithmic indicator following MMT level 3.9. Possible values: - 'ALGO': Algorithmic Trade - '-': No Algorithmic Trade Definitions: ALGO – transactions executed as a result of an investment firm engaging in algorithmic trading as defined in Article 4(1)(39) of Directive 2014/65/EU.
Filler	286	2	*Binary*	*For future use*

1.3.5 Auction Summary - 245 Message

1.3.5.1 Message Overview

An Auction Summary message summarizes an instrument's opening or auction trades. The Auction Summary message is sent by the trading engine after an instrument auction that has effectively traded, in order to summarize the auction, or after the first trade(s) if it/they occurred during continuous trading.

1.3.5.2 Message Sending Rules

This message indicates, for each security, the cumulative quantity traded during an auction/a first trade.

1.3.5.3 Message Structure

The table below describes the body fields of an Auction Summary message (MsgType = '245').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	245 - Auction Summary
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	Specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Example: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
FirstPrice	16	4	Binary Int.	First traded price for the current trading day (To be calculated with the PriceScaleCode)
LastPrice	20	4	Binary Int.	Most recent traded price for the current trading day (until the message is sent). (To be calculated with the PriceScaleCode)
HighestPrice	24	4	Binary Int.	Highest traded price for the current trading day (To be calculated with the PriceScaleCode)
LowestPrice	28	4	Binary Int.	Lowest traded price for the current trading day (To be calculated with the PriceScaleCode)
CumulativeQuantity	32	4	Binary Int.	Cumulative quantity traded for the current trading day (does not take into account hidden quantity)
Variation	36	4	Binary Int. (signed)	Percentage variation of the last trade price against the previous day's reference price for the instrument concerned, ie the last known price or the last price indication (variation price).(To be calculated with the VariationScaleCode).
SystemID	40	4	Binary Int.	The ID of the originating Exchange/System of the message
SourceTimeMicroSecs	44	2	Binary Int.	Number of micro seconds. To be combined with SourceTime.

Field Name	Offset	Size (Bytes)	Format	Description
TypeOfLastPrice	46	2	Binary Int.	Indicates the message trade type. Valid values: 4 - 1st traded price 7 - nth traded price (resume after unhalting of the instrument)
TickDirection	48	1	ASCII Ch.	Indicates how the last price has moved as compared to the preceding last traded price. Valid values: '+' - Rising '-' - Falling '0' - Unchanged or data field not significant (used for trade cancellation)
InstrumentValuationPrice	49	1	ASCII Ch.	Indicates whether the data item "last price" is a valuation price following a valuation trade Valid values: '0' - Not a valuation price '1' - Valuation price
PriceScaleCode	50	1	Binary Int.	Applicable to all prices in the message
VariationScaleCode	51	1	Binary Int.	Available for the price variation in the message

1.3.6 VWAP – Closing Price - 247 Message

1.3.6.1 Message Overview

This message contains the Volume-Weighted Average Price for an instrument, along with few statistical information related to the cumulated traded value and volume. It also transports the Closing Price (the closing price that can be the Last Traded Price or the VWAP).

1.3.6.2 Message Sending Rules

This message is sent at the End of Day – once the referential system batch has been performed – and provides the Last Adjusted Closing Prices for all financial instruments (taking into account the Corporate Actions that may have taken place).

1.3.6.3 Message Structure

The table below describes the body fields of a VWAP-Closing Price message (MsgType = '247').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	247 – VWAP / Closing Price Update
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day.

Field Name	Offset	Size (Bytes)	Format	Description
				Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SendingIndicator	16	2	Binary Int.	Indicates the "sending mode". Valid values: 1 - Real Time (not used for WSE) 2 - Batch mode.
ClosingPriceRule	18	2	Binary Int.	Indicates the Closing Price calculation rule. Valid values: 1 - LTP (Last Traded Price) 2 - VWAP (Volume Weighted Average Price) Not Provided for Derivatives
ClosingPrice	20	4	Binary Int.	Current VWAP/Closing Price (To be calculated with the ScaleCode) Not Provided for Derivatives
AdjustedClosingPrice	24	4	Binary Int.	Adjusted VWAP/Closing Price (To be calculated with the ScaleCode)
NbTrades	28	4	Binary Int.	Number of Trades executed on this Instrument since the beginning of the Trading Day
				Not Provided for Derivatives
AmountTraded	32	8	Binary Int.	Total cumulated amount traded on this instrument since the beginning of the Trading day (To be calculated with the ScaleCode) Not Provided for Derivatives
QtyShares	40	4	Binary Int.	Cumulated Quantity traded (number of securities) on this instrument since the beginning of the Trading Day Not Provided for Derivatives
SystemID	44	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
SourceTimeMicroSecs	48	2	Binary Int.	Number of micro seconds to be combined with SourceTime
ScaleCode	50	1	Binary Int.	Applicable to ClosingPrice, AdjustedClosingPrice and AmountTraded.
Filler	*51*	*5*	*Binary*	*Reserved for future use*

1.4 Quotes & BBO5 and IOI

1.4.1 Overview

The Quotes & BBO5 and IOI service uses the push-based publishing model. This means that data will be published based on its availability. Once a Quote is available, it will be published to clients.

The Quotes & BBO5 message reflects a configurable number of highest bids and lowest offers (e.g. BBO1, BBO5, BBO10 …) in each traded security.

Note: WSE will publish the 5 Best Bid and Offers (BBO5), however in the future this may be change to an other value.

List of the message types in the Equities Quotes Feed:

- 140 – Quotes & BBO5 - 52 Bytes in total, **50** excluding MsgSize field
- 142 – Indication Of Interest - 164 Bytes in total, **162** excluding MsgSize field

List of the message types in the BondSpot Quotes Feed:

- 140 – Quotes & BBO5 - 52 Bytes in total, **50** excluding MsgSize field

1.4.2 Quotes & BBO5 - 140 Message

1.4.2.1 Message Overview

A Quotes & BBO5 message indicates the modification of one or more of the N best limits (N being the number of limits disseminated for the given exchange[1]) for an instrument and, during the Pre-Opening phase, the modification of the "market summary" for the instrument.

[1] For WSE, the 5 Best Limits are disseminated.

1.4.2.2 Message Sending Rules

This message is sent:

- Each time one or more of the N best limits for an instrument is modified or when during the call (Pre-Opening) phase, the "market summary" is modified (then the message is sent with field QuoteNumber = 0).
- The "market summary" for an instrument is the summary of the orders that would be executed if the auction took place at the moment that this message was sent. It is meaningful only during a Call phase or when the instrument is in Halted mode w/ acceptation of orders, when an IMP is determined. The market summary is not provided if the IMP is equal to the first limit.

Note

The field SourceSeqNum contains the same message sequence number that is present in IAD Q message (field MsgSeqNum).

1.4.2.3 Message Structure

The table below describes the body fields of a Quotes & BBO5 message (MsgType = '140').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int..	140 – Quotes & BBO5
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message.

Field Name	Offset	Size (Bytes)	Format	Description
				Please note that while the sequence number increases serially, it does not increase monotonically.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170.
QuoteLinkID	16	4	Binary Int.	*Not used*
AskPrice	20	4	Binary Int.	Ask Price for Quote or BBO (To be calculated with the ScaleCode) Specific values: '0' - Limit Deletion 'FF FF FF FE ' - Market Order or Market To Limit Order
AskSize	24	4	Binary Int.	Total number of shares requested in sell orders at the ask price.
BidPrice	28	4	Binary Int.	Bid Price for Quote or BBO (To be calculated with the ScaleCode) Specific values: '0' - Limit Deletion 'FF FF FF FE ' - Market Order or Market To Limit Order
BidSize	32	4	Binary Int.	Total number of securities requested in buy orders at the bid price.
SystemID	36	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
NumberAskOrders	40	2	Binary Int.	Number of sell orders at the ask price.

Field Name	Offset	Size (Bytes)	Format	Description
				Set to 1 when QuoteNumber is equal to 0 (Market Summary)
NumberBidOrders	42	2	Binary Int.	Number of buy orders at the bid price Set to 1 when QuoteNumber is equal to 0 (Market Summary)
SourceTimeMicroSecs	44	2	Binary Int.	Number of micro seconds. To be combined with SourceTime.
TypeOfAskPrice	46	1	Binary Int.	Valid values: 0 - Limit order and Peg order 1 - Market order and Market To Limit order
TypeOfBidPrice	47	1	Binary Int.	Valid values: 0 - Limit order and Peg order 1 - Market order and Market To Limit order
QuoteCondition	48	1	ASCII Ch.	Liquidity provider presence indicator, only provided for instruments traded on a LP quote driven book: Valid values: '0' - No Liquidity provider (LP) '1' - LP only on Ask side '2' - LP only on Bid side '3' - LP on Ask and Bid sides Null - Not provided/Not applicable
QuoteNumber	49	1	Binary Int.	Indicates the level in the order book for the given aggregate quote, derived from it's price value.

Field Name	Offset	Size (Bytes)	Format	Description
				Level '0' is dedicated to the market summary (The market summary for an instrument is the summary of the orders that would be executed if the opening of an instrument took place at the moment when this message is sent – applicable only in Call mode when an IMP has been determined).
PriceScaleCode	50	1	Binary Int.	Applicable to all prices in the message
Filler	*51*	*1*	*Binary*	*For future use.*

1.4.3 Indication of Interest - 142 Message

1.4.3.1 Message Overview

The Indication of Interest message disseminates an indication of interest for a given instrument to all participants.

1.4.3.2 Message Sending Rules

This message is sent each time an, indication of interest is created, cancelled or expired.

1.4.3.3 Message Structure

The table below describes the body fields of an Indication of Interest message (MsgType = '142').

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Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 byte MsgSize field.
MsgType	2	2	Binary Int.	142 – Indication of Interest
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SourceTime	12	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
IOIUTPID	16	4	Binary Int.	IOI identifier: uniqueness is guaranteed for the day for all instruments.
IOIQty	20	4	Binary Int.	Optional, IOI quantity.
Price	24	4	Binary Int.	Optional, IOI price, to be calculated with PriceScaleCode
ValidityTime	28	4	Binary Int.	Optional, indicates the IOI end of validity time (HHMMSS). Time use CET.
SystemID	32	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
SourceTimeMicroSecs	36	2	Binary Int.	Number of micro seconds to be combined with SourceTime

Field Name	Offset	Size (Bytes)	Format	Description
PriceScaleCode	38	1	Binary Int.	Applicable to all prices in the message
IOITransType	39	1	ASCII Ch.	Indicates the IOI type. Valid values: 'N' - New 'C' - Cancel 'E' - Expire
FirmID	40	11	ASCII Ch.	Indicates the Firm ID who issued the indication of interest if the IOI is not anonymous
Side	51	1	ASCII Ch.	Indicates the side of the IOI. Valid values: '1' - Buy '2' - Sell
SettlementDate	52	8	ASCII Ch.	Optional, settlement date (YYYYMMDD)
Text	60	100	ASCII Ch.	Free text
TradeType	160	1	ASCII Ch.	IOI trade type. Valid values: 'O' - Block Trade 'S' - BISO Trade
Filler	*161*	*3*	*Binary*	*For future use*

1.5 Order Book

1.5.1 Overview

The OrderBook service uses the push-based publishing model. This means that data will be published based on its availability. Once information is available, it will be published to clients.

List of the messages in the WSE OrderBook:

- 230 - Order update - 64 Bytes in total, **62** excluding MsgSize field
- 231 - Orderbook retransmission delimiter (used when the Public OrderBook is entirely re-disseminated for one or several given instruments) - 16 Bytes in total, **14** excluding MsgSize field

List of the messages in the BondSpot OrderBook:

- 230 - Order update - 64 Bytes in total, **62** excluding MsgSize field

1.5.2 Order Update / Market Sheet - 230 Message

1.5.2.1 Message Overview

The Order update message generated by the trading engine UTP indicates the creation or modification of a line in the "market sheet" (a.k.a. "market depth" or "Public Order Book") of normal orders for an instrument. It is also used when the Market Sheet of normal orders is rebroadcast. The deletion of an order from the Market Sheet of orders is indicated via the ActionType fields 'D' of the Market Sheet message:

- This message takes into account any order type, except Stop Loss and Stop Limit (Stop orders).
- Stop orders are not broadcasted to the market participants until they are triggered.

1.5.2.2 Message Sending Rules

This message is sent:

- In the morning, when UTP is initialized, to retransmit orders remaining in the book from previous days (taking into account expired orders and order book purges). This is known as the 'orderbook retransmission' or 'market sheet retransmission' – action type - Y.
- During the day, each time an order introduced by a member firm modifies the market sheet (for example, when creating, modifying or changing the priority of an order), except during Auction processing.

To be noted:

- SymbolIndex+ OrderDate+ OrderID uniquely identifies an order
- Market sheet sequencing at Call Phase:
  - Orders must be arranged according to:
    1. Order type (OrderType): Priorities are first **Market** and **Market to limit**, second **Limit** orders.
    2. then Order price (Price)
    3. then Order priority timestamp (OrderPriorityDate+OrderPriorityTime+OrderPriorityMicroSecs)
- Market sheet sequencing at Continuous Trading Phase:
  - Orders must be arranged according to:
    1. Order type (OrderType): Priorities are first **Market**, second **Limit** orders and **Peg** orders.
    2. then Order price (Price)
    3. then Order priority timestamp (OrderPriorityDate+OrderPriorityTime+OrderPriorityMicroSecs)

1.5.2.3 Message Structure

The table below describes the body fields of an Order Update / Market sheet message (MsgType = '230').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	230 - Order Update / Market sheet
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the time when the Order Update is generated. Note, when an order is added, the SourceTime represents the order entry time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit. In the specific case of a Peg order repricing, the sequence number remains the same.
Price	16	4	Binary Int.	Price (Tto be calculated with the PriceScaleCode
AggregatedVolume	20	4	Binary Int.	Total quantity of orders at the given price level BondSpot: Not applicable
Volume	24	4	Binary Int.	Displayed quantity of the order. BondSpot: only applicable for ActionType 'A' and 'M'
LinkID	28	4	Binary Int.	*Not used*
OrderID	32	4	Binary Int.	Identifies the uniqueness of the order, when combined with OrderDate.
SystemID	36	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to possible values in appendix System ID.
SourceTimeMicroSecs	40	2	Binary Int.	Number of micro seconds. To be combined with SourceTime.
NumberOrders	42	2	Binary Int.	Number of orders in the Book at the price of the current order BondSpot: Not applicable
Side	44	1	ASCII Ch.	Indicates the side of the order. Valid values: '1' - Buy '2' - Sell '5' - Sell short, equivalent to value '2' (sell). Use of designation '5' as 'Sell short' by Exchange Members will be optional and will not be required or monitored by the WSE. Exchange Members may use the designation to mark orders with value '5' in the 'Side' field for its own needs. Null - Not provided
OrderType	45	1	ASCII Ch.	Type of Order. Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'1' - Market order '2' - Limit order 'K' - Market to limit order 'P' - Peg order Null - Not provided (case of order deletion independently of reason)
ActionType	46	1	ASCII Ch.	This field identifies the type of operation on order book. Values reserved for future use : 'O', 'C', 'E', 'X', 'Z' Valid values: 'A' - New order 'M' - Modification of existing order (case of partial fill) 'D' - Deletion of order identified by OrderID 'F' - Deletion all orders for one instrument at the start of the day before the order book retransmission 'Y' - Retransmission of all orders for the given instrument
PriceScaleCode	47	1	Binary Int.	Applicable to all prices in the message
OrderDate	48	4	Binary Int.	Date of order (YYYYMMDD). To be combined with OrderId.
OrderPriorityDate	52	4	Binary Int.	Date giving the priority of the order (YYYYMMDD)
OrderPriorityTime	56	4	Binary Int.	Time giving the priority of the order (HHMMSSsss)
OrderPriorityMicroSecs	60	2	Binary Int.	Number of micro seconds in the current millisecond BondSpot: Set to zero
OrderOrigin	62	1	ASCII Ch.	*Not used*
Filler	*63*	*1*	*Binary*	*For future use*

1.5.3 Order Book Retransmission Delimiter - 231 Message

1.5.3.1 Message Overview

The Orderbook retransmission delimiter message is set to clean then update the market sheet with the orders from previous days still remaining in the order book.

1.5.3.2 Message Sending Rules

This message is sent at the beginning of the trading session, and allows member to clean then update the market sheet with the orders remaining in the order book. A message is sent per instance of the trading engine (this is identified by the field InstanceNum).

1.5.3.3 Message Structure

The table below describes the body fields of an Orderbook Retransmission Delimiter message (MsgType = '231').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	231 - Orderbook retransmission delimiter Message
SourceTime	4	4	Binary Int.	This field specifies the time when the message is generated. Note, when an order is added, the SourceTime represents the order entry time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	8	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit.
ApplicationID	12	2	ASCII Str.	Code identifying the application where the orderbooks are retransmitted. Valid values: W0
InstanceNum	14	1	Binary Int.	Indicates the number of trading units for which the orderbooks are rebroadcast. (On the initial WSE configuration, the number of instances is 2 as there are 2 trading units) Valid values: 2
RetransmissionIndicator	15	1	ASCII Ch.	Indicates the status of the retransmission for the instance of the trading engine. Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'B' - Beginning of the retransmission 'E' - End of the retransmission

1.6 INDICES

1.6.1 Overview

The Indices service uses the push-based publishing model. This means that data will be published based on its availability. Once information is available, it will be published to clients.

List of the message types in the WSE Indices feed and the treasury BondSpot feed:

- 546 - Real Time Index - 112 Bytes in total, **110** excluding MsgSize field
- 547 - Index Summary - 104 Bytes in total, **102** excluding MsgSize field
- 548 - Index Composition - 704 Bytes in total, **702** excluding MsgSize field

1.6.2 Real time Index - 546 Message

1.6.2.1 Message Overview

The real time index message handles the real-time characteristics of an index.

1.6.2.2 Message Sending Rules

This message is sent:

- Before opening the session
- During the session
- At the end of the session

The message refers to the current information on index values, index associated indicators and other information products based on data aggregation. The subsequent levels and indicators of the phases appear along with the changes of the phases of individual indexes. The changes of the phases might be associated with the passage of time and the occurrence of the events, especially including exceeding the determined level of the Index Portfolio Opening Indicator W(t).

This description refers to:

- Resolution of the Management Board of the Warsaw Stock Exchange No. 42/2007 of January 16,2007 as amended – hereinafter referred to as the WSE Index Resolution,
- Resolution of the Management Board of the Warsaw Stock Exchange No. 452/2007 of July 13,2007 as amended – hereinafter referred to as the NewConnect Index Resolution,
- Resolution of the Management Board of the Warsaw Stock Exchange No. 651/2009 of November 18,2009 as amended – hereinafter referred to as the RESPECT Index Resolution,
- Regulations of the Treasury BondSpot Poland Index hereinafter referred to as the TBSPIndex Regulations.

1.6.2.3 Message Structure

The table below describes the body fields of a Real Time Index message (MsgType = '546').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	546 - Real Time Index

Field Name	Offset	Size (Bytes)	Format	Description
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170.
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
IndexLevel	20	4	Binary Int.	The value of the last level for the index according to the definition provided in the IndexLevelCode (To be calculated with the LevelScaleCode). Refer to the Index Appendix. For the levels '8', '3', '9' and 'X' the field is completed with zero.
SessionHigh	24	4	Binary Int.	Index highest level of the day (To be calculated with the LevelScaleCode)
				(not including the opening index value at the level 'A' and the average weighted index with level 'R'). The field is completed for the massages with level '1', '2' and '5 during index status „ZW" and „ZO"
SessionLow	28	4	Binary Int.	Index lowest level of the day (To be calculated with the LevelScaleCode) (not including the opening index value at the level 'A' and the average weighted index with level 'R'). The field is completed for the massages with level '1', '2' and '5 during index status „ZW" and „ZO"
IndIndexOpeningPortfolio	32	4	Binary Int.	Indicator of the index opening portfolio W(t) (To be calculated with the LevelScaleCode) The percentage change of the value of the packages of the instruments, for which the quotation was set to the total value of the index portfolio. During the index status „TO" indicator W(t) includes the set Theoretical Opening Quotations of the instruments. The field is completed with zero for messages with the level 'X', 'Y', 'R' and '5' during the index status „ZI".

Field Name	Offset	Size (Bytes)	Format	Description
				For strategy indices (in 548 message the field ProductClassification filled in with value “S”, “L” or “M”) the field is filled in with value zero.
PercentChangeTOI	36	4	Binary Int.	Percentage change of TOI in relation to the value of the closing index from the previous session (To be calculated with the LevelScaleCode) The field is completed only for messages with the level ‘8’, ‘3’ and ‘9’.
PctChangeIndexValPrevSession	40	4	Binary Int.	Percentage change of the index value in relation to the value of the closing index from the previous session (To be calculated with the LevelScaleCode) The field is completed for messages with level ‘1’, ‘A’, ‘2’, ‘C’ and ‘5’during the index status „ZW” and „ZO”. Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
PctChangeIndexValPrevYear	44	4	Binary Int.	Percentage change of the index value in relation to its value at the end of the previous year (To be calculated with the LevelScaleCode) The field is completed for messages with level ‘5’during the index status „ZW” and „ZO”. Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
TOI	48	4	Binary Int.	Theoretical opening index calculated during the index status „TO” (the field completed only for message with level ‘8’) or during the index status „PO” (the field completed only for message with level ‘3’ and ‘9’). (To be calculated with the LevelScaleCode) Note: The field completed depending on the value of the indicator in message 548 in field TOIDissemination.„Publication of the theoretical opening index (TOI) during the index status TO”.
PointChange	52	4	Binary Int.	The point change of the index to the previous value of the index (To be calculated with the LevelScaleCode). The field is completed for messages with level ‘2’.
ValueTrading	56	8	Binary Int.	Value of trading in the instruments from the given index (To be calculated with the LevelScaleCode). The field is completed for messages with level ‘3’, ‘9’, ‘1’, ‘2’ and ‘5’ during the index status „ZW” and „ZO” for selected indexes. Note: the field is completed depending on the value of the indicator in message 548 in field TradingValuePublication„The indicator of the publication of the trading volume”.

Field Name	Offset	Size (Bytes)	Format	Description
IndexValueBid	64	4	Binary Int.	The calculated value of the buy index based on the best buy offers of instruments (To be calculated with the LevelScaleCode) The field is completed for messages with level '3', '1' and '2' till the end continuous trading phase. Note: the field is completed depending on the value of the indicator in message 548 in field Spread Dissemination „Publication of spread indicators".
IndexValueAsk	68	4	Binary Int.	The calculated value of the sell index based on the best sell offers of instruments. (To be calculated with the LevelScaleCode) The field is completed for messages with level '3', '1' and '2' till the end continuous trading phase. Note: the field is completed depending on the value of the indicator in message 548 in field Spread Dissemination „Publication of spread indicators
MidSpreadIndex	72	4	Binary Int.	Arithmetic mean of the values of the fields: "Index value according to buy offers" and „Index value according to sell offers" (To be calculated with the LevelScaleCode) The field is completed for messages with level '3', '1' and '2' till the end continuous trading phase.
				Note: the field is completed depending on the value of the indicator in message 548 in field SpreadDissemination„ Publication of spread indicators".
DifferenceCentralSpread	76	4	Binary Int.	The point difference of central spread to the current value of index (To be calculated with the LevelScaleCode) The field is completed for messages with level '1' and '2' till the end continuous trading phase. Note: the field is completed depending on the value of the indicator in message 548 in field SpreadDissemination„Publication of spread indicators"
NumOfActiveInstruments	80	2	Binary Int.	Number of instruments – participants of the index, for which the quotation was set during the session. The field is completed with zero for the messages with level: 'X', 'Y', 'R' and '5' during the index status „ZI". For strategy indices (in 548 message the field ProductClassification filled in with value "S", "L" or "M") the field is filled in with value zero.
NumOfInstrOfDecreasingQt	82	2	Binary Int.	The number of instruments from the index portfolio, whose quotation at the current session is lower than the reference quotation at the session on the previous day.

Field Name	Offset	Size (Bytes)	Format	Description
				The field is completed for messages with level '1', 'A', 'B', 'C', and '5' during the index status „ZW" and „ZO". For strategy indices (in 548 message the field ProductClassification filled in with value "S", "L" or "M") the field is always filled in with value zero. Possible values: The value in the field lower or equal to the number of instruments comprising the index portfolio. In special cases the value can be 0.
NumOfInstrOfIncreasingQt	84	2	Binary Int.	The number of instruments the from index portfolio, whose quotation at the current session is higher than the reference quotation at the session on the previous day. The field is completed for messages with level '1', 'A', 'B', 'C', and '5'during the index status „ZW" and „ZO". For strategy indices (in 548 message the field ProductClassification filled in with value "S", "L" or "M") the field is always filled in with value zero. Possible values: The value in the field lower or equal to the number of instruments comprising the index portfolio. In special cases the value can be 0.
NumOfInstrOfUnchangedQt	86	2	Binary Int.	The number of instruments from the index portfolio, whose quotation at the current session did not change as compared with the closing quotation at the session on the previous day. For strategy indices (in 548 message the field ProductClassification filled in with value "S", "L" or "M") the field is always filled in with value zero. The field is completed for messages with level '1', 'A', 'B', 'C', and '5'during the index status „ZW" and „ZO". Possible values: The value in the field lower or equal to the number of instruments comprising the index portfolio. In special cases the value can be 0.
SourceTimeMicroSecs	88	2	Binary Int.	Number of micro seconds to be combined with SourceTime
LevelScaleCode	90	1	Binary Int.	To be combined with values of the following fields : IndexLevel SessionHigh, SessionLow, IndIndexOpeningPortfolio, PercentChangeTOI, PctChangeIndexValPrevSession, PctChangeIndexValPrevYear, TOI, PointChange, ValueTrading, IndexValueBid, IndexValueAsk, MidSpreadIndex, DifferenceCentralSpread

Field Name	Offset	Size (Bytes)	Format	Description
TimeFirstHigh	91	6	ASCII Ch.	The time at which the highest value of a given index was published for the first time. The field is completed for the messages with level '1', '2', '5' and 'Y'. Time format: HHMMSS. Time use CET. Note: For the messages with level 'Y' the field assumes the value of time, for which the index was calculated at the session. The value is consistent with the time interval of publishing a given index.
TimeFirstLow	97	6	ASCII Ch.	The time at which the lowest value of a given index was published for the first time. The field is completed for messages with level '1', '2' and '5' during index status „ZW" and „ZO".Format HHMMSS. Time use CET.
IndexStatusIndicator	103	2	ASCII Ch.	Two letter marker informing about the index status at the time of publishing the message. Valid values: 'ZI' - index status „Index setting"; during this phase, the message B1 with level '5' is sent with the index closing value from the previous session 'TO' - index status „Theoretical Opening"; during this phase, the message B1 with level '8' is sent
				'PO' - index status „Before Opening"; during this phase, the message B1 with level '3' and '9' is sent 'SI' - index status „Session"; during this phase, the messages B1 with levels 'A', '1', '2' and 'B' are sent 'ZW' - index status „Preliminary Closing"; during this phase, the messages B1 with levels 'C', '5' and 'R' and message B2 are sent. (the message with level 'R' is not published at each session and refers to the special applications of the WSE). 'IM' - index status „Index Manager Intervention"; during this phase, the messages B1 with levels 'X' are sent (if required), and then with levels 'Y' 'ZO' - index status „Final Closing "; during this phase, the messages B1 with levels 'C','5' and 'R' and message B2 are sent again. (the message with level 'R' is not published at each session and refers to the special applications of the WSE).
IndexLevelCode	105	1	ASCII Ch.	Index Level Code. Valid values (for the full levels description, refer to the Index Appendix) : '1' - First value of the index at the session '2' - Current index '3' - Theoretical value of the index before its opening

Field Name	Offset	Size (Bytes)	Format	Description
				'5' - Closing index from previous session (reference index) '8' - Theoretical opening index before opening the session '9' - Index without opening 'A' - Preliminary opening index 'B' - Synthetic opening index 100% 'C' - Final opening index 'R' - Average weighted index 'X' - Cancellation of the current values of the index 'Y' - Recalculated current values of the index
MarkerPercentChangeTOI	106	1	ASCII Str.	Marker for the percentage change of TOI in relation to the value of the closing index from the previous session. The field is completed only for messages with the level '8', '3' and '9'.Valid values: '+' - Increase '-' - Decrease Space - No change
MarkerPctChgIndexValPrevSess	107	1	ASCII Str.	The indicator for the point change of the current value of the index as compared with the previous index value. The field is completed for messages with level '2'. Valid values: '+' - Increase '-' - Decrease Space - No change Not used for strategy index (in 548 message the field ProductClassification filled in with value "M")
MarkerPctChgIndexValPrevYear	108	1	ASCII Str.	Marker for the percentage change of the index value in relation to its value at the end of the previous year. The field is completed for messages with level '5'during the index status „ZW" and „ZO". Valid values: '+' - Increase '-' - Decrease Space - No change Not used for strategy index (in 548 message the field ProductClassification filled in with value "M")
MarkerPointChange	109	1	ASCII Str.	Marker for the point change of the index to the current value of the index. Valid values: '+' - Increase '-' - Decrease Space - No change
MarkerDifferenceCentralSpread	110	1	ASCII Str.	The marker for the point change between the central spread and the last published index value. The field is completed for messages with level '1' and '2' till the end continuous trading phase.

Field Name	Offset	Size (Bytes)	Format	Description
				Valid values: '+' - Increase '-' - Decrease Space - No change
DisseminationFrequencyMarker	111	1	ASCII Ch.	One letter marker determining, whether the information on index value will be published in the next message at a particular level. Valid values: '0' - When the message at a particular level is published regularly at the session '1' - When the message at a particular level is published at the session on a one time basis

1.6.3 Index Summary - 547 Message

1.6.3.1 Message Overview

The index summary message provides a summary of the data calculated in a stock index for a given day.

1.6.3.2 Message Sending Rules

This message is sent for every index at the end of the session, when the index closing values are calculated and approved.

This description refers to:

- Resolution of the Management Board of the Warsaw Stock Exchange No. 42/2007 of January 16,2007 as amended – hereinafter referred to as the WSE Index Resolution,
- Resolution of the Management Board of the Warsaw Stock Exchange No. 452/2007 of July 13,2007 as amended – hereinafter referred to as the NewConnect Index Resolution,
- Resolution of the Management Board of the Warsaw Stock Exchange No. 651/2009 of November 18,2009 as amended – hereinafter referred to as the RESPECT Index Resolution,
- Regulations of the Treasury BondSpot Poland Index hereinafter referred to as the TBSPIndex Regulations.

1.6.3.3 Message Structure

The table below describes the body fields of Index Summary message (MsgType = '547').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	547 – Index Summary
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170.
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message.

Field Name	Offset	Size (Bytes)	Format	Description
				In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
ClosingIndexValue	20	4	Binary Int.	Closing index value (To be calculated with the LevelScaleCode). Value of closing index from the last message 546 with level „5”.
IndOpeningPortfolio	24	4	Binary Int.	Indicator of the opening of the index portfolio W(t) for closing index. The percentage change of the value of the packages of instruments, for which the quotations were set at the session to the total value of index portfolio. For strategy indices (in 548 message the field ProductClassification filled in with value “S”, “L” or “M”) the field is filled in with value zero. (To be calculated with the LevelScaleCode).
OpeningIndexValue	28	4	Binary Int.	Opening index value (To be calculated with the LevelScaleCode). = opening index value at a particular session from level „C” of message 546
SessionHigh	32	4	Binary Int.	Highest level of the day (To be calculated with the LevelScaleCode). It does not include the index values calculated according to synthetic formula from levels A, B and C from message 546.
SessionLow	36	4	Binary Int.	Lowest level of the day (To be calculated with the LevelScaleCode) It does not include the index values calculated according to synthetic formula from levels A, B and C from message 546.
PctChgHighIndexValPrevSession	40	4	Binary Int.	Percentage change of the highest index value to the value of the closing index from the previous session (To be calculated with the LevelScaleCode) Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
PctChgLowIndexValPrevSession	44	4	Binary Int.	Percentage change of the lowest index value to the value of the closing index from the previous session (To be calculated with the LevelScaleCode) Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)

Field Name	Offset	Size (Bytes)	Format	Description
PctChangeIndexValPrevSession	48	4	Binary Int.	Percentage change of the closing index value in relation to the value of the closing index from the previous session (To be calculated with the LevelScaleCode) Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
PctChangeIndexValPrevYear	52	4	Binary Int.	Percentage change of the index value in relation to its value at the end of the previous year (To be calculated with the LevelScaleCode) Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
CapitalisationPortfolio	56	4	Binary Int.	Capitalisation of the portfolio for the closing index expressed in the currency of the index, on basis of which the closing index is calculated. For strategy indices (in 548 message the field ProductClassification filled in with value “S” or “L”) the field is filled in with value of closing underling index. (to be combined with CapitalisationScaleCode) Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
IndexAdjustmentCoefficient	60	4	Binary Int.	Index adjustment coefficient K(t) for the closing index (To be combined with IndexAdjustmentScaleCode)
				The algorithm of setting its value is provided in the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations For strategy indices (in 548 message the field ProductClassification filled in with value “S” or “L”) the field is filled in with value zero. Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
SourceTimeMicroSecs	64	2	Binary Int.	Number of micro seconds to be combined with SourceTime
LevelScaleCode	66	1	Binary Int.	To be combined with Indices values (ClosingIndexValue, IndOpeningPortfolio, OpeningIndexValue, SessionHigh, SessionLow, PctChgHighIndexValPrevSession, PctChgLowIndexValPrevSession, PctChangeIndexValPrevSession, PctChangeIndexValPrevYear)
CapitalisationScaleCode	67	1	Binary Int.	To be combined with CapitalisationPortfolio Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
IndexAdjustmentScaleCode	68	1	Binary Int.	To be combined with IndexAdjustmentCoefficient
TimeClosing	69	6	ASCII Ch.	Time of the publication of the closing value of the index at a given session.. Format HHMMSS. Time use CET.

Field Name	Offset	Size (Bytes)	Format	Description
TimeOpening	75	6	ASCII Ch.	Time of the publication of the opening value of the index. The value is equal to the time from level „A" of message 546 or if there is no message 546 with level „A" then from message 546 with level „C". Format HHMMSS or zero in case the publication time is not defined in detail. Time use CET.
TimeFullOpening	81	6	ASCII Ch.	Time of the publication of the full opening value of the index portfolio for W(t) = 100%. The time of sending the message 546 with level 'B' Format HHMMSS or zero in case the indicator W(t) has not reached 100%. Time use CET.
TimeHigh	87	6	ASCII Ch.	Time of the publication of the highest value of the index. Format HHMMSS. Time use CET.
TimeLow	93	6	ASCII Ch.	Time of the publication of the lowest value of the index. Format HHMMSS. Time use CET.
MarkerHigh	99	1	ASCII Str.	Marker of the percentage change of the highest value of the index to the closing index value from the previous session. Valid values: '+' - Increase '-' - Decrease Space - No change
				Not used for strategy index (in 548 message the field ProductClassification filled in with value "M")
MarkerLow	100	1	ASCII Str.	Marker of the percentage change of the lowest value of the index to the closing index value from the previous session. Valid values: '+' - Increase '-' - Decrease Space - No change Not used for strategy index (in 548 message the field ProductClassification filled in with value "M")
MarkerClosingPrevSession	101	1	ASCII Str.	Marker of the percentage change of the index at a particular session to the closing index value from the previous session. Valid values: '+' - Increase '-' - Decrease Space - No change Not used for strategy index (in 548 message the field ProductClassification filled in with value "M")
MarkerClosingPrevYear	102	1	ASCII Str.	Marker of the percentage change of the closing value of the index to its value at the end of the previous year. Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'+' - Increase '-' - Decrease Space - No change Not used for strategy index (in 548 message the field ProductClassification filled in with value "M")
Filler	*103*	*1*	*Binary*	*For future use.*

1.6.4 Index Portfolio - 548 Message

1.6.4.1 Message Overview

Index portfolio.

1.6.4.2 Message Sending Rules

This message is sent:

- Before the session
- During the session

This message is sent for the indexes and other information products published by the Warsaw Stock Exchange.

This description refers to:

- Resolution of the Management Board of the Warsaw Stock Exchange No. 42/2007 of January 16,2007 as amended – hereinafter referred to as the WSE Index Resolution,
- Resolution of the Management Board of the Warsaw Stock Exchange No. 452/2007 of July 13,2007 as amended – hereinafter referred to as the NewConnect Index Resolution,
- Resolution of the Management Board of the Warsaw Stock Exchange No. 651/2009 of November 18,2009 as amended – hereinafter referred to as the RESPECT Index Resolution,
- Regulations of the Treasury BondSpot Poland Index hereinafter referred to as the TBSPIndex Regulations.

1.6.4.3 Message Structure

The table below describes the body fields of an Index Portfolio message (MsgType = '548').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	548 - Index Portfolio
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the index itself
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit.

Field Name	Offset	Size (Bytes)	Format	Description
				In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message. Refer to appendix System ID for possible values.
IndexAdjustmentCoefficient	20	4	Binary Int.	Index adjustment coefficient K(t) for the index (To be combined with IndexAdjustmentScaleCode) The algorithm of setting its value is provided in the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations For strategy indices (in 548 message the field ProductClassification filled in with value “S” or “L”) the field is filled in with value zero. Not used for strategy index (in 548 message the field ProductClassification filled in with value “M”)
LevelCashPortfolio	24	4	Binary Int.	Level in cash in portfolio For strategy indices (in 548 message the field ProductClassification filled in with value “S” or “L”) the field is filled in with value of interbank reference rate adopted in the index calculation algorithm. (To be combined with CashPortfolioScaleCode)
ValueLastClosingIndex	28	4	Binary Int.	Value of the last closing index / NAV value number of ETF units (To be combined with LevelScaleCode)
NumberOfComponents	32	2	Binary Int.	Total number of instruments in the index / given product For strategy indices (in 548 message the field ProductClassification filled in with value “S” or “L”) the field is filled in with number of days between current and previous session. For strategy index (in 548 message the field ProductClassification filled in with value “M”) filled in with the number of dividends included in the index calculation
SourceTimeMicroSecs	34	2	Binary Int.	Number of micro seconds to be combined with SourceTime
IndexAdjustmentScaleCode	36	1	Binary Int.	To be combined with IndexAdjustmentCoefficient
CashPortfolioScaleCode	37	1	Binary Int.	To be combined with LevelCashPortfolio
LevelScaleCode	38	1	Binary Int.	To be combined with ValueLastClosingIndex
Beginning of characteristics of index portfolio / information product; block is repeated 20 times				
InstrumentISIN	39	12	ASCII Ch.	ISIN code of the instrument
InstrumentAbbrevName	51	10	ASCII Ch.	Abbreviated name of the instrument
InstrumentPacket	61	10	ASCII Ch.	Instrument packet in index portfolio / of information product Number of units (usually stocks/shares) of a given instrument in the index portfolio or in the information product. For strategy indices (the field ProductClassification filled in with value “S”, “L” or “M”) the field is not filled in.

Field Name	Offset	Size (Bytes)	Format	Description
End of characteristics of index portfolio / information product				
DateValidity	679	8	ASCII Ch.	Date of validity of index portfolio / information product (YYYYMMDD).
IndexCurrency	687	3	ASCII Ch.	Index currency / information product currency
IndexStatusIndicator	690	2	ASCII Ch.	Index Status Indicator. Valid values: 'PS' - Preparation of the new session"; during this index status 548 message is prepared, which is sent during „ZI" index status. 548 message can be prepared and sent during one of the following phases in the case of changing the portfolio during the course of a particular session (during the phase other than „PS" index status):'ZI' - Index setting 'TO' - Theoretical opening 'PO' - Before opening 'SI' - Session 'ZW' - Preliminary closing 'IM' - Index manager intervention 'ZO' - Final closing
IndexPresSeq	692	2	ASCII Ch.	Index presentation sequence marker For other information product the field :filler
ModeIndexDissemination	694	1	ASCII Ch.	Mode of index dissemination.
				One letter marking advising how the index is published. The publication method is stipulated under the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations. Valid values: 'C' - Index is quoted in determined time intervals 'F' - Index is quoted on determined hours For other information product the field: filler
ProductClassification	695	1	ASCII Ch.	Information product classification. Valid values: 'W' - WSE index from the main market 'Z' - External index 'V' - Another WSE information product 'M' - WSE strategy dividend point index 'S' - WSE strategy index of short type 'L' - WSE strategy index of leverage type
OpeningCalcFormula	696	1	ASCII Ch.	Opening index calculation formula. One letter marker informing, whether for a particular index the synthetic (on the first transactions quotations) or chronological (on the current quotations) value of opening the index is set. Valid values:

Field Name	Offset	Size (Bytes)	Format	Description
				'S' - Synthetic 'O' - Chronological For other information product the field: filler
Filler	*697*	*1*	*Binary*	*For future use.*
TOIDissemination	698	1	ASCII Ch.	TOI dissemination marker during TO index status. Valid values: 'Y' - Message 546 with level 8 is sent 'N' - Message 546 with level 8 is not sent For other information product the field: filler
SpreadDissemination	699	1	ASCII Ch.	Spread dissemination marker. Valid values: 'Y' - Spread indicators are published 'N' - Spread indicators are not published
TradingValuePublication	700	1	ASCII Ch.	Trading value publication marker. One letter marker informing whether the value of the trading in the instruments from a particular index portfolio is provided for a particular index (in the message 546 – field ValueTrading). Valid values: 'Y' - Value is provided 'N' - Value is not provided For other information product the field: filler
MessageContinuation	701	1	ASCII Ch.	Message Continuation Marker. Valid values: '0' - The message is not the last message information of the composition of the index portfolio '1' - The message is the last message information of the composition of the index portfolio
Filler	*702*	*2*	*Binary*	*For future use.*

1.7 BOND SPECIFIC MESSAGES

1.7.1 Overview

The Bond service uses the push-based publishing model. This means that data will be published based on its availability. Once information is available, it will be published to clients.

List of the messages for WSE:

- 601 – Accrued interest - 276 Bytes in total, **274** excluding MsgSize field
- 602 – Interest for settlement day - 44 Bytes in total, **42** excluding MsgSize field

List of the messages for BondSpot:

- 601 – Accrued interest - 276 Bytes in total, **274** excluding MsgSize field
- 602 – Interest for settlement day - 44 Bytes in total, **42** excluding MsgSize field
- 603 – Yield to maturity - 48 Bytes in total, **46** excluding MsgSize field

1.7.2 Accrued Interest - 601 Message

1.7.2.1 Message Overview

This message informs about the accrued interest for bonds and mortgage-backed bonds. The field "NumberDays" defines the number of filled group fields "AccruedInterestGroup".

1.7.2.2 Message Sending Rules

This message is sent:

- In the morning at the start of the session.
- During the day, each time the value of interest for a given bond or mortgage-backed bond is changed.

1.7.2.3 Message Structure

The table below describes the body fields of a BondSpot Accrued Interest message (MsgType = '601').
The combination "Date/Accrued Interest" is repeated 31 times.

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	601 – Accrued Interest
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message

Field Name	Offset	Size (Bytes)	Format	Description
NumberDays	20	2	Binary Int.	Number of days for which accrued interest is provided. Valid values: From 1 to 31.
SourceTimeMicroSecs	22	2	Binary Int.	Number of micro seconds to be combined with SourceTime
AccruedInterestGroup	24 mod (8)			Group repeated 31 times:
Date		4	Binary Int.	Date of accrued interest (YYYYMMDD). Provided with 0 if date is not significant
AccruedInterest		4	Binary Int.	Accrued interest for the day defined by field date. Provided with 0 if date is not significant
AccruedInterestScaleCode	272	1	Binary Int.	Applicable to all accrued interests
Filler	*273*	*3*	*Binary*	*For future use*

1.7.3 Interest for Settlement Day - 602 Message

1.7.3.1 Message Overview

This message informs about the interest for settlement day for bonds and mortgage-backed bonds.

1.7.3.2 Message Sending Rules

This message is sent:

- In the morning at the start of the session;
- During the day, each time the value of interest for settlement day, for a given bond or mortgage-backed bond is changed.

1.7.3.3 Message Structure

The table below describes the body fields of a BondSpot Interest For Settlement Day message (MsgType = '602').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	602 – Interest for Settlement Day
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message
Date	20	4	Binary Int.	Stock Exchange date (YYYYMMDD).

Field Name	Offset	Size (Bytes)	Format	Description
SettlementDate	24	8	ASCII Ch.	Settlement date (YYYYMMDD). Provided with 0 if settlement date is not significant.
Interest	32	4	Binary Int.	Interest for the settlement day defined by field "SettlementDate". Provided with 0 if settlement date is not significant.
SourceTimeMicroSecs	36	2	Binary Int.	Number of micro seconds to be combined with SourceTime
Currency	38	3	ASCII Str.	Code of the currency in which the interest is determined. (ISO 4217-3A)
InterestScaleCode	41	1	Binary Int.	Applicable to the interest for the settlement day
Filler	*42*	*2*	*Binary*	*For future use.*

1.7.4 Yield to Maturity - 603 Message

1.7.4.1 Message Overview

This message informs about the values of particular types of yield to maturity (YTM). Message is sent for treasury bonds and bills only for BondSpot.

Note

603 message is distributed together with: 140, 241 messages with the field "price index" equal '50','02'or'03', 240 – for TBS Poland market.

1.7.4.2 Message Sending Rules

This message is sent:

- After the determination of fixing price on a fixing of treasury BondSpot Poland
- After each trade
- After an update of one of the best five limits (buy or sell)
- In case of 140 (Quotes & BB05) messages occurred:
  - The 603 message is sent for each price level changing, separately for bid and ask price, in case of both are sent within 140,
  - The 603 messages are sent right after the package of 140 concerning the same event is sent.

1.7.4.3 Message Structure

The table below describes the body fields of a Yield To Maturity (YTM) message (MsgType = '603').

Field Name	Offset	Size (Bytes)	Format	Description
MsgSize	0	2	Binary Int.	Length of the message body, excluding the 2 bytes MsgSize field.
MsgType	2	2	Binary Int.	603 – Yield to maturity
SymbolIndex	4	4	Binary Int.	UTP-MD proprietary identification of the instrument
SourceTime	8	4	Binary Int.	This field specifies the message generation time. The number in this field represents the number of milliseconds since midnight of the same day. Ex: If SourceTime = 13:12:56 secs, 170ms and 30 microsecs, this field will contain 47576170
SourceSeqNum	12	4	Binary Int.	This field specifies the sequence number assigned by the source system to this message. In Real Time mode, the sequence number increases serially and monotonically for the same Trading Unit. In RFS mode, the sequence number of message does not increases serially and does not increase monotonically for the same Trading Unit.

Field Name	Offset	Size (Bytes)	Format	Description
SystemID	16	4	Binary Int.	The ID of the originating Exchange/System of the message
Date	20	4	Binary Int.	Stock Exchange date (YYYYMMDD).
Price	24	4	Binary Int.	Indicates the value of the price depending on the indicator in field MessageIndicator. Valid values: - fixing price, for which the YTM value is calculated in the field "YTM" - bid (indicative buy price) – price of instrument calculated in accordance with the Treasury Securities Fixing Rules determined by NBP (National Bank of Poland) - ask (indicative sell price) – price of instrument calculated in accordance with the Treasury Securities Fixing Rules determined by NBP(National Bank of Poland) - trade price, price of trade for which the YTM value is specified in the field "YTM" - buy limit – buy quote limit for the relevant level (one of the first five best limits depending on the field "Message level") at the time of sending the message - sell limit — sell quote limit for the relevant level (one of the first five best limits depending on the field "Message level") at the time of sending the message - filled with '0' in case of undetermined fixing for the instrument
SourceTimeMicroSecs	28	2	Binary Int.	Number of micro seconds to be combined with SourceTime
YTMScaleCode	30	1	Binary Int.	Applicable to the YTM
PriceScaleCode	31	1	Binary Int.	Applicable to the price
YTM	32	4	Binary Int.	Indicates the values of the Yield To Maturity depending on the indicator in field MessageIndicator. Valid values: - YTM value of fixing price – yield to maturity rounded off to 1 basis point calculated according to the formula attached in the Exhibit to the Treasury Securities Fixing Rules determined by NBP. (National Bank of Poland) - Fixing YTM value for bid (indicative buy price) – YTM value corresponding to the bid (indicative buy price) at the settlement date according to the KDPW timetable (D+ 2). - Fixing YTM value for ask (indicative sell price) – YTM value corresponding to the ask (indicative sell price) at the settlement date according to the KDPW timetable (D+ 2). - YTM value for a trade – the current YTM value for the price of a last trade (YTM can be the negative value) - YTM for buy limit – YTM corresponding to the given buy best limit level (depending on the field" Level of the message" (YTM can be the negative value) - YTM for sell limit – YTM corresponding to the given sell best limit level (depending on the field" Level of the message" (YTM can be the negative value) - filled with '0' in case of undetermined fixing for the instrument
MessageIndicator	36	1	ASCII Str.	Indicates the value of the fields 'Price' and 'YTM'

Field Name	Offset	Size (Bytes)	Format	Description
				Valid values: 'F' - fixing 'T' - trade 'K' - bid (indicative buy price) or the best buy quite price (depending on the field "Level of the Message") 'S' - ask (indicative sell price) or the best sell quote price (depending on the field "Level of the Message") The way of filling in the fields „Price" and „YTM" depending on the MessageIndicator" value and" "MessageLevel", was described in the table provided in Appendix §2.7 (YTM).
AlgorithmIndicator	37	1	ASCII Str.	Field includes information about the methodology of calculating YTM. Valid values: '1' - ISMA/LLP algorithm '2' - ISMA algorithm
MaturityIndicator	38	1	ASCII Str.	Valid values: '0' - date of maturity in accordance with a prospect '1' - previously date of maturity
MessageLevel	39	1	ASCII Str.	Valid values: 0 to 5. If the field contains the value '0', the fields "Price" and "YTM" contain relevant values depending on the value of field "Message indicator".

Field Name	Offset	Size (Bytes)	Format	Description
				If the field contains the values: '1', '2', '3','4' or '5', the field indicates for which level of the buy or sell price limit (one of the first five best limits) the YTM is indicated in the field "YTM".
MaturityDate	40	4	Binary Int.	Maturity date (YYYYMMDD).
TradeIDNumber	44	4	Binary Int.	If the field "Message indicator" = 'T', it is the unique trade number for the given instrument during the given trading session. Otherwise the field is filled in with zeros.

2 APPENDICES

2.1 STOCK TYPES

Code	Stock type	Product Sub Type
000	Index	
100	Equity	Ordinary Shares
101	Equity	Preferred Shares
102	Equity	Preference Shares
103	Equity	Convertible Shares
104	Equity	Depositary Receipts
105	Equity	Unit Trust Units
106	Equity	Investment Trust Shares
107	Equity	National Fund Certificates
108	Equity Linked;	Allotment Rights
109	Equity Linked	Subscription Rights
110	Equity Linked	Issue Rights
111	Structured Product	Cash Certificates
112	Equity Linked;	Subscription Warrants
200	Bond	Bonds
201	Bond	Treasury Bonds
202	Bond	Treasury Notes
203	Bond	Treasury Bills
204	Bond	Municipal Bond
205	Bond	Corporate Bonds
206	Bond	Commercial Papers
207	Bond	Convertible Bonds
208	Bond	Mortgage-Backed Bonds
209	Bond	Public Mortgage Bonds
210	Structured Product	Cash Structured Bonds
300	Tracker	ETF
301	Tracker	ETN
302	Tracker	ETC
303	Tracker	ETV
400	Structured Product	Warrants
401	Structured Product	Warrants with Knock-Out
402	Structured Product	Certificates with Knock-Out
403	Structured Product	Leveraged Structured Certificates
404	Structured Product	Certificates
410	Structured Product	Structured Bonds
500	Futures	Futures
600	Options	Vanilla Options
601	Options	Binary Options
700	Participation Unit	Index Participation Units

2.2 SYSTEM ID

ID	Application
1	All non UTP systems
11	UTP Trading Unit 1
12	UTP Trading Unit 2

2.3 STOCK EXCHANGE CODE

ID	Application
243	Warsaw Stock Exchange
986	BondSpot Non Exchange Regulated Market
988	Treasury BondSpot Poland
989	BondSpot ATS

2.4 SECTORS

2.4.1 Sector Classification for the main Market Segments of WSE

New sector classifications of issuers were update accordingly as of the beginning of 2017. The new rules apply to issuers of shares listed on both WSE regulated market and alternative market.

Value	Description
111	commercial banks
112	co-operative banks
113	mortgage banks
114	government and international banks
119	banks - other
121	insurance offices
129	insurance - other
131	exchanges and brokers
132	asset management
139	capital market - other
141	real estate sales
142	real estate rent
149	real estate - other
150	Leasing & Factoring
160	Mortgage
170	Financial Services
180	Investment
190	Financials - Other
211	oil & gas exploration and production
212	oil & gas refining and transportation
219	oil & gas - other
221	power engineering
222	eco-energy
229	energy - other
230	Water & Multi-utilities
290	Oil & Energy - Other
311	basic chemicals
312	specialty chemicals
319	chemicals - other
321	carbon mining
322	metals mining
329	mining - other
331	steel metallurgy
332	non-ferrous metallurgy
339	metallurgy - other
340	Basic Materials
351	plastic
352	rubber
359	rubber & plastic - other
361	forestry
362	paper & packing
369	paper & forestry - other
370	Recycling
390	Chemicals & Materials - Other
411	general construction
412	industry construction

Value	Description
413	engineering
414	construction materials
415	building products
419	construction - other
421	electro machinery
422	industrial machinery
423	metals products
424	farm & heavy trucks
429	machinery - other
431	transport
432	logistics
439	transportation - other
441	electro products
442	steel products
443	chemical products
449	Industrial supplies - other
450	Industry Services
490	Industrials - Other
511	food
512	drinks
513	agricultural products and fishing
519	food & drinks - other
521	clothes & footwear
522	cosmetics & household chemicals
529	clothes & cosmetics - other
531	household durables
532	electrical & electronic equipment

Value	Description
539	consumer durables - other
541	auto parts
549	automobiles & parts - other
590	Consumer Goods - Other
611	consumer staples
612	computers & electronics
619	wholesale trade - other
621	consumer discretionary
622	consumer electronics
629	general retailers - other
631	hotels & restaurants
632	travel agencies
633	sports clubs
634	sport & leisure
639	leisure facilities - other
641	radio & TV
642	publishing
643	advertising
644	internet
649	media - other
650	Video Games
660	E-Trade
690	Trade & Services - Other
710	Health Care Services
720	Health Care Equipment
730	Pharmaceuticals
740	Pharmaceuticals Wholesales

Value	Description
750	Biotechnology
790	Health Care - Other
810	Telecom
821	software
822	IT systems
823	hardware
829	IT - other
830	New Technology
890	Technology - Other
910	State
921	voivodship
922	county
923	county state
924	community
929	local government - other
930	Central Banks
990	Government - Other
000	Non-Classified

2.5SEGMENTS

Value	Definition
LA	Segment Alerts' List
NF	Segment NewConnect Focus
NB	Segment NewConnect Base
NA	Segment NewConnect Alert

2.6INDICES

The field IndexLevelCode allows to determine the status and phase of the index .
The messages with level ‘8’, ‘3’ ,‘2’, ‘Y’ are sent regularly. The messages with the level ‘1’, ‘9’, ‘A’, ‘B’, ‘C’, ‘5’, ‘R’ and ‘X’ are sent upon the fulfilment of the conditions including especially the conditions stipulated under the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations.
The field assumes one of the following values:

Start of data transmission –index status „ZI ” (for all indexes)

- „5” – „closing index from previous session” (reference index), the value calculated on the basis of the last quotations at the previous session, and if such were not present on the basis of the reference quotations; the message provided once at the beginning of the day before starting the market session during index status “ZI” of the index.

Index status „TO”

- „8” – "Theoretical Opening Index before opening the session”, the message is sent continuously (regularly) at time intervals determined for a particular index. The message is sent after the message with level ‘5’, from the hour determined by the WSE to the end of the index status ‘TO’ of the index. Depending on the parameterisation of the index the message the message can be sent or not. The message quotes the Theoretical Opening Index (TOI) and the associated indicators (in the fields ‘Indicator for the percentage change of the index during the phase before its opening’ and ‘Percentage change of the index at the phase before its opening’) on the basis of TOP of instruments.

During the session – index status „PO”

- „3” – "Theoretical value of the index before its opening", The message quotes the value of the index before its opening and the associated indicators (in the fields ‘Indicator for the percentage change of the index during the phase before its opening’ and ‘Percentage change of the index at the phase before its opening’) on the basis of the current quotations of the instruments; the messages with this level are sent from the time of entering into the first transaction in the instrument from the portfolio of a particular index to the time of fulfilling the conditions stipulated under the WSE Index . Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations. The condition of starting the publication is the achievement by the Index Portfolio Opening Indicator W(tgiven-value (the quotient of the capitalisation of packages of the instruments for which the quotations were set during the session to the time of the complete capitalisation of the portfolio for a particular index).
- „9” – "index without opening", message is sent for indexes calculated under the uniform system on the determined hour in the situation, when the conditions stipulated under the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations are not met (usually when the Portfolio Opening Indicator W(t) has not exceeded the given value).

During the session - index status „SI”

- „A” – "preliminary opening index " set depending on the parameterisation of the index stipulated under the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations For the opening value according to the synthetic formula the index is calculated on the basis of the quotations of first transactions, and under chronological formula on the basis of the current quotations from the time of start of publishing the current values of the index; message with level ‘A’ is sent immediately before message 546 with level ‘1’.
- „1” – "the first value of the index at the session", the value calculated on the basis of the quotations of the last transactions; the message is sent after meeting the conditions stipulated under the WSE Index

Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations (after the achievement the given by indicator W(t)) or time. The message 546 with level 'A' is published before this message.

- „2" – "current index", the value calculated on the basis of the current quotations; the message is sent at the time intervals determined for a particular index. For example for the continuous indexes:

WIG20	– every 15 seconds
WIG20short	– every 15 seconds
WIG20lev	– every 15 seconds
mWIG40	– every 1 minute
sWIG80	– every 1 minute
WIG30	– every 1 minute
WIG	– every 1 minute
WIGdiv	– every 1 minute
RESPECTIndex	– every 1 minute
NCindex	– every 5 minutes

For strategy indices (in 548 message the field ProductClassification filled in with value "S" or "L") they are published according to the publication timetable of underlying indices.

For the uniform indexes with level '2' it can be sent once according to the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations.

The first message at this level is sent only after publishing the messages 546 with level 'A' and '1'.

- „B" – "synthetic opening index 100%", the message is sent only for the indexes, whose opening value is calculated according to the synthetic formula – the index is calculated on the basis of the quotations of the first transactions in individual instruments after reaching the value of 100% by the indicator W(t)); The message cannot be sent, when during the session the condition W(t) = 100% is not met.

For indexes calculated under the uniform system (for example indexes: WIG-Poland, TBSPIndex) the hours of publishing messages 546 are stipulated in detail. Depending on the phase of the index, there can be messages with various levels of indexes consistent with the descriptions of the indexes under a continuous system. Only the messages 546 with level '8', '3' and 'B' are not present. The messages with the level '8' and '3' for uniform indexes are replaced with the message with level '9'.

At the end of the session for all indexes – index status „IM".

- "X" – „cancellation of the current values of the index" – technical level informing about cancelling the previously quoted values of the index (messages published during a particular session). After the message with this level the correct recalculated values of a particular index at a particular session shall be published - starting from publishing the first value of the index.
- "Y" – „recalculated current values of the index" – the messages are preceded with the publication of the message with level „X", they replace the messages with level '1', '2' and '5'(from index status 'ZW'), contain only the current values of the indexes and the time, for which they were recalculated. The messages are sent between publishing the first and second message with level „5" at the end of the stock market session.

At the end of the session for all indexes – index status „ZW, ZO".

- „C" – "the final opening index" set depending on the parameterisation of the index stipulated under the WSE Index Resolution, NewConnect Index Resolution, RESPECT Index Resolution and TBSPIndex Regulations. For the opening values under synthetic formula – the index is calculated on the basis of the first quotations of the transactions and under chronological formula on the basis of the current quotations (then it is equal to the index value from level 'A'). The message is sent twice during the index status „ZW" and „ZO" in association with the message with level '5'. Sending the message during index status „ZO" means „the approved value of the opening index ".
- „5" – "closing index ", message sent during the index status ZW and ZO of the index; the value is calculated on the basis of the quotations of the last transactions at a given session; the message is sent twice: the first one with level '5' is sent during ZW index status and is a „preliminary closing index". The second message with level '5' is sent during ZO index status and is the „approved closing index". The values of the index at this level might differ.
- „R" – „average weighted index" – the value calculated on the basis of opening indexes and from the time interval determined by the index manager. The message is not published during each session and refers to special applications of the WSE or BondSpot. The message is sent twice during the index status „ZW" and „ZO" in association with the publication of the messages with level „C" and „5", Sending the message during the index status „ZO" means „the approved value of the average weighted index ".

2.7YTM

Level of Messages	Message indicator	Field	Value of the Field
0	F	Price	Fixing price
		YTM	Fixing price YTM
0	T	Price	Trade price
		YTM	Ttrade price YTM
0	K	Price	Bid (indicative buy price)
		YTM	Fixing YTM value for bid (indicative buy price)
0	S	Price	Ask (indicative sell price)
		YTM	Fixing YTM value for ask (indicative sell price)
1	K	Price	First level buy limit
		YTM	YTM for first level buy limit
1	S	Price	First level sell limit
		YTM	YTM for first level sell limit
2	K	Price	Second level buy limit
		YTM	YTM for second level buy limit
2	S	Price	Second level sell limit
		YTM	YTM for second level sell limit
3	K	Price	Third level buy limit
		YTM	YTM for third level buy limit

Level of Messages	Message indicator	Field	Value of the Field
3	S	Price	Third level sell limit
		YTM	YTM for third level sell limit
4	K	Price	Fourth level buy limit
		YTM	YTM for fourth level buy limit
4	S	Price	Fourth level sell limit
		YTM	YTM for fourth level sell limit
5	K	Price	Fifth level buy limit
		YTM	YTM for fifth level buy limit
5	S	Price	Fifth level sell limit
		YTM	YTM for fifth level sell limit

2.8MIC CODES

	Country	MIC	Institution Description
1	POLAND	XWAR	Warsaw Stock Exchange
2	POLAND	XNCO	Warsaw Stock Exchange/ Equities/New Connect-MTF
3	POLAND	WBON	Warsaw Stock Exchange/ Bonds/Catalyst/Main Market
4	POLAND	WMTF	Warsaw Stock Exchange/Bonds/Catalyst/MTF
5	POLAND	WDER	Warsaw Stock Exchange/Financial Derivatives
6	POLAND	WETP	Warsaw Stock Exchange/ ETPs
7	POLAND	WIND	Warsaw Stock Exchange/Indices
8	POLAND	RPWC	Warsaw Stock Exchange/Bonds/Catalyst/Bondspot/Main Market
9	POLAND	BOSP	Warsaw Stock Exchange/Bonds/Catalyst/Bondspot/MTF
10	POLAND	TBSP	Warsaw Stock Exchange/Bonds/Bondspot/Treasury Bond Market
11	POLAND	WBCL	Warsaw Stock Exchange/ Bonds/Catalyst/Listing
12	POLAND	WIPO	Warsaw Stock Exchange/Other than XOFF or XXXX
13	ZZ	XOFF	OFF-Exchange Transactions – Listed instruments
14	ZZ	XXXX	No market (e.g. Unlisted)

Trading Class	ARM	MIC	Description
I1 - I2 - Primary market I3 - I4 - Secondary market	Yes	XOFF	1 - Financial instruments which are admitted to trading or traded on a trading venue or for which a request for admission to trading has been made where transactions in such financial instruments are not executed on the trading venue, through a systematic internaliser or on an organized trading platform outside the EU or where the investment firm is not aware that it is trading with another investment firm acting as a systematic internaliser.
I5 - I6 - Primary market I7 - Secondary market	Yes	XXXX	2 - Financial instruments which are not admitted to trading or traded on a trading venue or for which no request for admission to trading has been made and which are not traded on an organized trading platform outside the EU where the underlying is admitted to trading or traded on a trading venue.
I8 - Primary market I9 - Secondary market	No	WIPO	3 - Financial instruments for which no request for admission to trading has been made and which do not meet the criteria of the point no 2 above.